

Reviewed Preprint

v1 • September 12, 2025

Not revised

Reviewed Preprint

v2 • January 16, 2026

Revised by authors

For correspondence:

Lan.Guan@ttuhsc.edu

Competing interests:

Y. S. and R. V. are employees of Thermo Fisher Scientific.

Funding:

See page 22

Reviewing editor:

Andres Jara-Oseguera, The University of Texas at Austin, United States

© 2025, Hariharan et al. This article is

distributed under the terms of the

[Creative Commons Attribution](#)

[License](#), which permits unrestricted

use and redistribution provided that

the original author and source are

credited.

Allosteric effects of the coupling cation in melibiose transporter MelB

Parameswaran Hariharan¹, Yuqi Shi², Bakhtiar Amirhossein³, Ruibin Liang³, Rosa Viner², Lan Guan¹ ✉

¹Department of Cell Physiology and Molecular Biophysics, Center for Membrane Protein Research, School of Medicine, Texas Tech University Health Sciences Center, Lubbock, United States • ²Thermo Fisher Scientific, San Jose, United States • ³Department of Chemistry and Biochemistry, Texas Tech University, Lubbock, United States

eLife Assessment

This manuscript presents **useful** insights into the molecular basis underlying the positive cooperativity between the co-transported substrates (galactoside sugar and sodium ion) in the melibiose transporter MelB. Building on years of previous studies, this **convincing** study improves on the resolution of previously published structures and reports the presence of a water molecule in the sugar binding site that would appear to be key for its recognition, introduces further structures bound to different substrates, and utilizes binding and transport assays, as well as HDX-MS and molecular dynamics simulations to further understand the positive cooperativity between sugar and the co-transported sodium cation. The work will be of interest to biologists and biochemists working on cation-coupled symporters, which mediate the transport of a wide range of solutes across cell membranes.

<https://doi.org/10.7554/eLife.108335.2.sa3>

Abstract

The major facilitator superfamily (MFS) transporters play significant roles in human health and disease. *Salmonella enterica* serovar Typhimurium melibiose permease (MelB_{St}), which catalyzes the symport of galactosides with Na⁺, H⁺, or Li⁺, is a prototype of this important transporter superfamily. We have published the structures of the inward- and outward-facing conformations of MelB_{St} with galactoside or Na⁺ bound, determined the binding thermodynamic cycle, and proposed that positive cooperativity between the two co-transported solutes plays a key role in the symport mechanism of MelB_{St}. The molecular basis for this core mechanism remains unclear. In this study, we determined the molecular basis for this core symport mechanism through analyzing the structural dynamics of MelB_{St} and effects induced by melibiose, Na⁺, or both using hydrogen-deuterium exchange mass spectrometry (HDX-MS). We also refined the specific determinants for the sugar recognition in both protein and galactoside molecules by solving the crystal structures of a uniporter D59C MelB_{St} bound to melibiose and other sugars, and identified a critical water molecule as part of sugar recognition. Our integrated studies from structure, HDX-MS, and molecular dynamics simulations support the conclusion that sugar-binding affinity is directly correlated with protein dynamics. The binding of the coupling cation at a remote site functions as an allosteric activator to restrain the conformational flexibility of dynamic residues in the sugar-binding site and in the cytoplasmic gating salt-bridge network, thereby increasing sugar-binding affinity allosterically. This study provides a molecular-level schematic of the fundamental symport mechanism via positive cooperativity, which may serve as a general mechanism for cation-coupled symporters.

Introduction

The Solute Carrier (SLC) family of transporters encompasses diverse superfamilies of membrane proteins with various protein folds and employing different transport mechanisms to facilitate the translocation of a wide range of solutes across cell membranes (1). The largest superfamily of SLC transporters is the major facilitator superfamily (MFS)(2), which includes a significant number of cation-coupled secondary active transporters. MFS transporters are responsible for the uptake of a broad spectrum of solutes across cell membranes, playing crucial roles in physiology, pathology, and pharmacokinetics, and are emerging as drug targets (3, 4). Recent rapid advancements in membrane protein research have greatly enhanced our understanding of protein conformation and mechanisms(5–8); however, critical details remain lacking. For example, in cation-coupled symport, it is still unclear how the coupling cations facilitate the binding, translocation, and accumulation of the primary substrate.

For MFS secondary active transporters, most members use H^+ as the coupling cation, and a few members use Na^+ , such as the Na^+ -coupled essential lipid transporter (MFSD2A) that is expressed in the major organ barriers, including the blood-brain barrier or blood-retina barrier (9–11). The Na^+ -coupled melibiose transporter of *Salmonella enterica* serovar Typhimurium (MelB_{St}), which is a well-characterized representative for the Na^+ -coupled MFS transporters, catalyzes the symport of a galactopyranoside with Na^+ , H^+ , or Li^+ , and is a valuable model system for studying cation-coupled transport mechanisms (12–22). Two major conformations of MelB_{St} have been determined: an outward-facing conformation at the apo or galactoside-bound states(20, 22) and an Na^+ -bound inward-facing conformation (21). The primary substrate-specificity determinant pocket and the cation-specificity determinant pocket have been structurally and functionally characterized (19–23). All three coupling cations compete for the same binding pocket, and the transport stoichiometry is 1 galactoside: 1 cation (Na^+ , H^+ , or Li^+). Binding of the primary and coupling substrates is positively cooperative; the sugar affinity depends on the cation identity, with the cooperativity numbers (fold of increase in affinity) being 8, 5, or 2 for Na^+ , Li^+ , and H^+ , respectively (19, 20). In addition to being sensitive to the binding of the coupling cation and its identity, notably, the sugar-binding affinity is also dependent on MelB_{St} conformation (21). By trapping MelB_{St} in an inward-facing state using the inward-facing conformation-specific binder nanobody-725 (Nb725), both experimental sugar-binding assays and cryoEM structural analysis support that the sugar-binding pocket at inward-facing conformation is at a low-affinity state (21, 23). Remarkably, Na^+ binding to the inward-facing conformation remains unchanged (21, 23). These results provide experimental evidence supporting the previously proposed stepped-binding kinetic model for melibiose/ Na^+ symport, in which Na^+ binds first and is released after the sugar release on the opposite surface (15, 18, 24, 25).

Positive cooperativity of substrate binding has been proposed to be the key symport mechanism in MelB_{St}, but the molecular basis for this critical mechanism remains unclear. The structures indicate that the bound sugar and Na^+ have no direct contact, while the two binding pockets are in close proximity (20, 21). The minimum free-energy landscape for sugar translocation, which was simulated based on the structures of the outward- and inward-facing conformations as two starting points (26), suggests that the Na^+ contribution to the binding free energy of sugar by direct contact is negligible. Cooperativity occurs through allosteric coupling, likely via electrostatic interactions.

In this study, we further analyzed the sugar-binding site by improving the crystal structure resolution of α -NPG bound at 2.60 Å, which uncovered an important water molecule in the binding site. We also confirmed the sugar specificity determinants in both sugar and MelB_{St} by determining the crystal structures in complex with three other α -sugar substrates—melibiose, raffinose, or α -methyl galactoside (α -MG)—that vary in the number of sugar units but all contain an α -galactosyl group. We then examined the structural dynamics of the entire MelB with hydrogen-deuterium exchange coupled to mass spectrometry (HDX-MS) and the effects of melibiose, Na^+ , or both on MelB_{St}. HDX-MS is particularly attractive for dynamic systems, as the provided information yields beyond the stable events within the dynamic repertoire of

transporter proteins. By monitoring the HDX rate across a time window, it can provide predictive modeling for overall protein folds (27, 28). The resolution of structural information provided by this technique is typically on the peptide level, though. We also performed MD simulations to analyze the side-chain dynamics. Collectively, all results support the notion that the sugar-binding affinity of MelB_{St} is coupled to protein structural dynamics and conformational transitions between inward- and outward-facing states, and the coupling cation in this symporter functions as an allosteric activator.

Results

Sugar transport and binding affinity measurements

The sugar analog α -NPG has been determined as a substrate for MelB_{Ec} by measuring *p*-nitrophenol production from the cells that expressed both MelB_{Ec} and α -galactosidase (29). The same assay was modified to determine the α -NPG translocation mediated by MelB_{St} in DW2 cells. Melibiose at 1 mM was added to induce α -galactosidase expression during cell growth, and the washed cells were applied to measure the time-course of *p*-nitrophenol release into the media upon adding 1 mM α -NPG into the induced cells. The detection was for the *p*-nitrophenol, which was resulted from α -NPG transport and hydrolysis. The results showed that both WT MelB_{St} and D59C uniport mutant mediated the translocation of α -NPG (Fig. 1a) at this downhill mode of transport.

To determine raffinose transport activity, [³H]raffinose transport in the *E. coli* DW2 strain was carried out in the absence of Na⁺ and Li⁺ or the presence of Na⁺ or Li⁺ (Fig. 1b). The results showed that raffinose, a trisaccharide formed from melibiose and fructose, is also a substrate and transported by MelB_{St}.

Previously, our ITC studies in the presence of Na⁺ revealed that the binding affinity, *K*_d values, of the WT and D59C MelB_{St} for melibiose were 1.25 mM $\pm$ 0.05 mM and 4.96 $\pm$ 0.11 mM, respectively, and for α -NPG were 16.46 $\pm$ 0.21 μ M or 11.97 $\pm$ 0.09 μ M, respectively (19, 20, 25, 30). The same assay was utilized to determine the α -MG and raffinose binding in the presence of Na⁺ (Fig. 1c). By injecting the α -MG or raffinose in a buffer-matched solution into the sample cell containing the WT MelB_{St} or the D59C uniporter mutant in the presence of Na⁺, the isotherm curve fitted well, yielding *K*_d values of 0.88 $\pm$ 0.04 mM or 3.79 $\pm$ 0.53 mM, for α -MG or raffinose, respectively. Raffinose, which has more sugar units, exhibits a poor binding affinity to MelB_{St}.

Outward-facing crystal structures of MelB_{St} complexed with varied sugar substrates

The uniporter D59C MelB_{St} mutant exhibits greater thermostability, making it a valuable tool for structural analysis of sugar binding (20). Here we report four crystal structures of D59C MelB_{St} with the endogenous sugar substrate melibiose, and two other α -galactosides, α -methyl galactoside (α -MG) with a single sugar unit or raffinose with three sugar units, as well as α -NPG at an improved resolution. The structure statistics were presented in Table S1. All structures adopt a virtually identical outward-facing conformation with RMSD values of less than 0.4 (Fig. S1). This typical MFS-fold transporter with 12 transmembrane helices is organized into two six-helix domains linked by the middle loop between helices VI and VII (Loop₆₋₇). There are three cytoplasmic helices at the loop₆₋₇, loop₈₋₉, and the C-terminal tail or lid (Fig. 2), named ICH1-3, respectively; and ICH1 and ICH2 run parallel to the membrane bilayer. In all structures, one sugar molecule is bound in the middle of the protein and sandwiched by both N- and C-terminal domains as described previously (20). As indicated by the sliced surface, the binding residues located within the cytoplasmic leaflet of both domains define the edge of the inner barrier, which prevents the sugar from passing across the transporter into the cytoplasm. On the periplasmic side, the open vestibule connects the solvent to the binding pocket. In the α -NPG-bound structure,

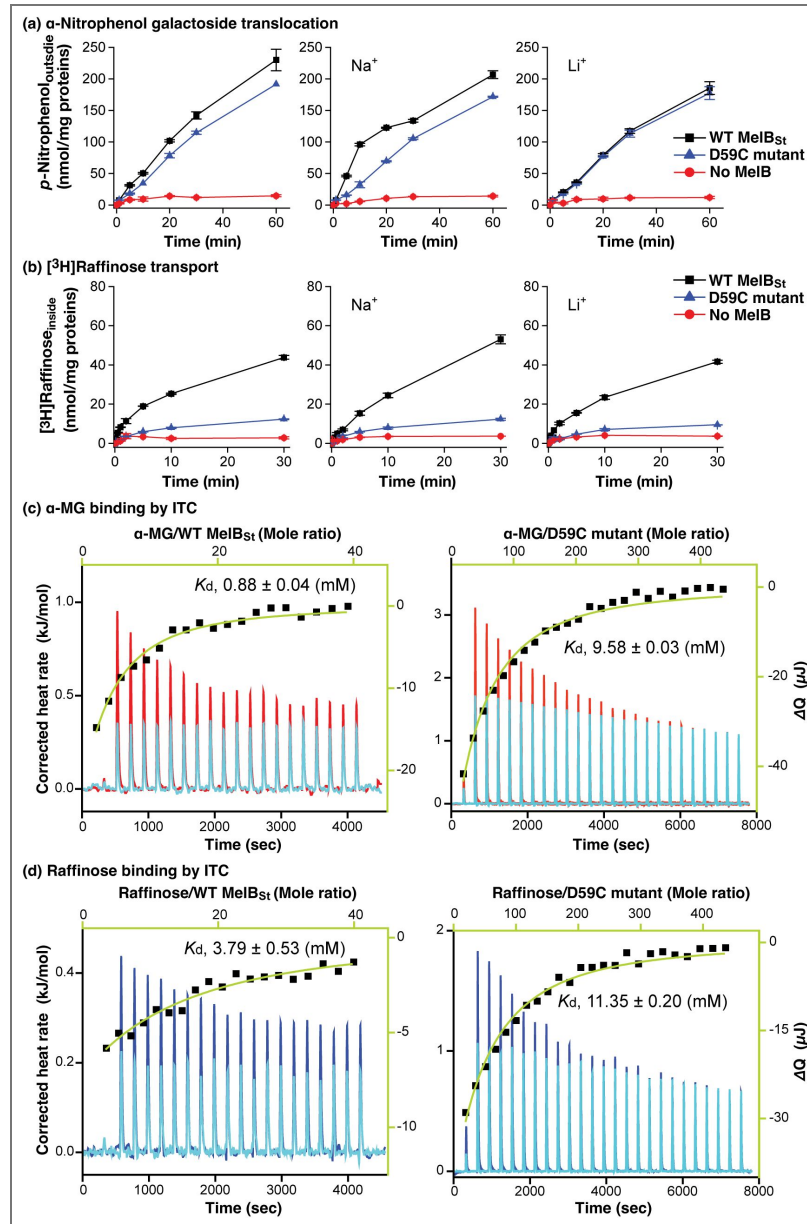


Fig 1. Functional characterizations.

(a) α-NPG down-hill transport. *E. coli* DW2 cells with the induced α-galactosidase by melibiose in the absence or presence of WT MelB_{St} or D59C mutant were washed before incubating with 0.5 mM *p*-nitrophenyl-α-D-galactoside at 30 °C in the absence or presence of 20 mM NaCl or 20 mM LiCl as described in Methods. The cell-aliquots at given time points at 0, 1, 5, 10, 20, 30, and 60 min were quenched with 0.3 M Na₂CO₃, followed by centrifugation to remove the cells. The *p*-nitrophenol in supernatant released from the cells was measured at A_{405} nm. **(b)** [3H]Raffinose active transport. The *E. coli* DW2 cells in the absence or presence of MelB_{St} with no α-galactosidase induction were used for the active transport of [3H]raffinose at 1 mM (specific activity, 10 mCi/mmol) at 23 °C in the absence or presence of 50 mM NaCl or LiCl. The cellular uptake time course measurements at 0, 0.08, 0.17, 0.5, 1, 2, 5, 10, and 30 min were carried out by a dilution and fast-filtration method. **(c & d)** ITC measurement of α-MG (c) or raffinose (d). ITC measurements were performed at 25 °C under similar buffer conditions: 20 mM Tris-HCl, pH 7.5, 100 mM NaCl, 10% glycerol, and 0.035% UDM detergent. For each experiment, 80 μM of the purified WT MelB_{St} or D59C mutant was placed in the reaction cell, and methyl α-D-galactoside (α-MG) or raffinose at 10 mM (against the WT) or 100 mM (against D59C) from the syringe are incrementally titrated to generate the thermograms. The curve fitting is performed with a one-site independent-binding model included in the NanoAnalyze software (version 3.7.5). The thermograms were plotted as baseline-corrected heat rate (μJ/sec; left axis) vs. time (bottom axis) for the titrant to MelB_{St} (red for α-MG and blue for raffinose) or to buffer (light blue). The heat change ΔQ (μJ; filled black symbol) was plotted against the mole ratio of the sugar to WT MelB_{St} (top/right axes in green). **(c)** α-MG. **(d)** Raffinose.

as reported in PDB ID 8FRH (D59C apo structure) or ID 8FQ9 (D55C mutant with DDMB(22)), a PEG molecule was modeled to the density near Arg296 along with helix IX, which could indicate a potential lipid-binding site (Fig. S1).

α -NPG-binding structure refined to a resolution of 2.60 Å

The improved resolution, from 3.01 Å to 2.60 Å, provided a better-resolved density map for the bound α -NPG molecule, which further supported the originally assigned pose, as demonstrated by the stereo view (Fig. 3 [↗](#)) (20). Thus, the binding pocket is formed by 14 residues on five helices, including N-terminal residues, as labeled in black including helices I (Lys18, Asp19, Ile22, and Tyr26), IV (Tyr120, Asp124, and Tyr128), and V (Arg149 and Ala152), and the C-terminal residues labeled in blue including helices X (Trp342) and XI (Gln372, Thr373, Val376, and Lys377). A water molecule (Wat 1) was modeled to a positive density, which located at hydrogen-bonding distances from the C4-OH and C6-OH on the galactopyranosyl ring and the Thr373 at helix XI, and also surrounded by Gln372 at helix XI and Asp124 at helix IV, as shown in the stereo-view of a 2Fo-Fc electron density map (Fig. 3 [↗](#)). Another three water molecules participated the cytoplasmic gating salt-bridge network between the both bundles (Fig. S2a, b). Water molecules Wat 2 and Wat 3 interacted with the charged pair Arg295 (helix IX) and Asp351 (helix XI), respectively, and Wat 4 interacted with Lys138 from loop₄₋₅, which also formed a salt-bridge interaction with Glu142 in this functionally important gating area (Fig. S2b) (31).

Melibiose binding

The endogenous substrate melibiose is formed from a galactose unit and a glucose unit linked by an α -1,6 galactosyl bond (D-Gal-(α 1 $\rightarrow$ 6)-D-Glc). The melibiose-bound D59C MelB_{St} structure was refined to a resolution of 3.05 Å (Fig. 4a, [↗](#)), and the density map displayed a clear two-unit blob, fitting well with one molecule of melibiose. Two water molecules, Wat 2 and Wat 3, were modeled; however, no strong peak was observed at the Wat 1 position. Notably, the binding affinity between melibiose and α -NPG differs by a factor of 100; in α -NPG, the hydrophobic phenyl group, positioned at 4 Å distance from the phenyl group of Tyr26, forms a strong stacking interaction. While melibiose has four additional OH groups, the glucosyl ring sits nearly perpendicular to the phenyl ring of Tyr26 (Fig. S3) with no strong polar interaction with the transporter.

Methyl α -D-galactoside (α -MG) binding

α -MG has a methyl substituent at the anomeric C1 position of the galactosyl moiety in an α -linkage. The crystal structure of D59C MelB_{St} complexed with α -MG was refined to a resolution of 3.68 Å, and the density map clearly displayed a one-unit blob in the binding pocket, where one α -MG was modeled similarly to the galactosyl moiety of melibiose (Fig. 4b [↗](#)).

Raffinose binding

The raffinose-bound D59C MelB_{St} mutant structure was refined to a resolution of 3.40 Å, and the density map displayed a clear three-unit blob in the binding pocket, fitting well with this trisaccharide. As expected, the raffinose-binding pocket is significantly larger than that of melibiose and α -NPG (Fig. 4c [↗](#)), involving two additional helices. Consequently, a set of polar side chains on helix VII (Asn244, Ser247, Asn248, and Asn251), helix VIII (Asn279), and helix V (Ser153) were within 4-5 Å distances to either the glucosyl or fructosyl moieties. Interestingly, the polar interactions from raffinose to MelB_{St} are limited to the galactosyl moiety, which are nearly identical to those presented in α -MG, melibiose, or α -NPG, while the Asp19 is at a hydrogen-bonding distance from the OH-4 on the glucosyl moiety.

All four galactosyl moieties align well in the specificity-determinant pocket, regardless of the number of monosaccharide units (Fig. 4d, [↗](#); Fig. S1), which provided a strong structural support to the previous conclusion that the galactosyl moiety determines the specificity of the primary substrates and the non-galactosyl moiety contributes to the binding affinity.

Fig. 2. Crystal structures of D59C MelB_{St} in complex with α -sugars substrates.

Upper row: Cartoon representation of the structures of D59C MelB_{St} bound with α -NPG, melibiose (α -disaccharide), and raffinose (α -trisaccharide), respectively, along with a surface presentation of D59C MelB_{St} complexed with α -methyl galactoside (α -MG). All structures were oriented with the cytoplasmic side on top and the N-terminal domain (colored green) on the left. Each sugar molecule is colored yellow. The blue sticks and surfaces indicate residues within 5 Å of the sugar molecules. **Lower row:** Sugar-binding pocket. Residues from the N-terminal and C-terminal domains were shown in surface representation, and colored in green and blue, respectively.

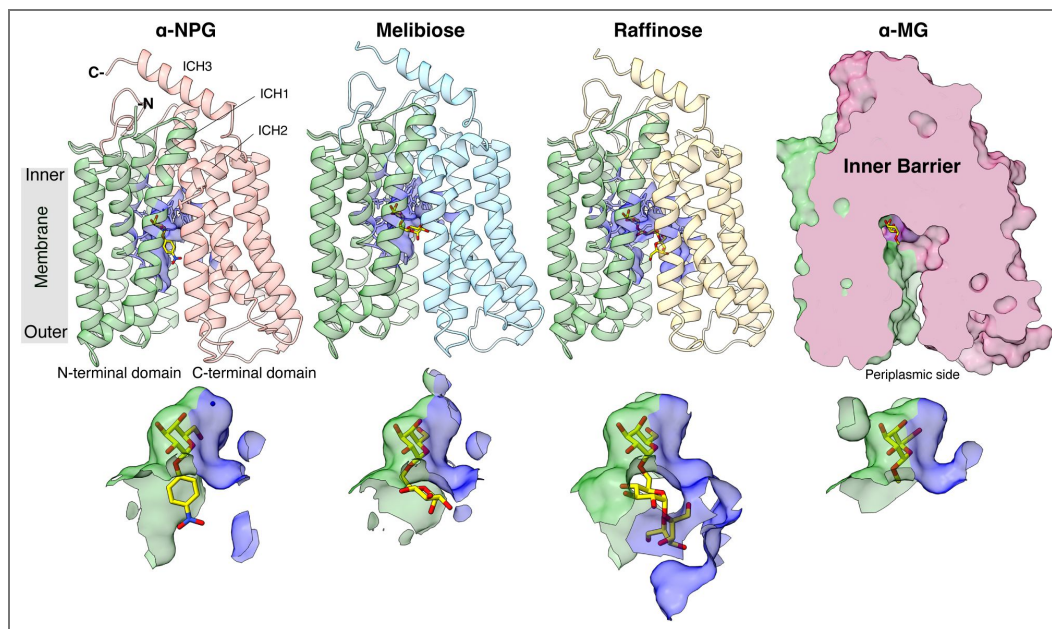
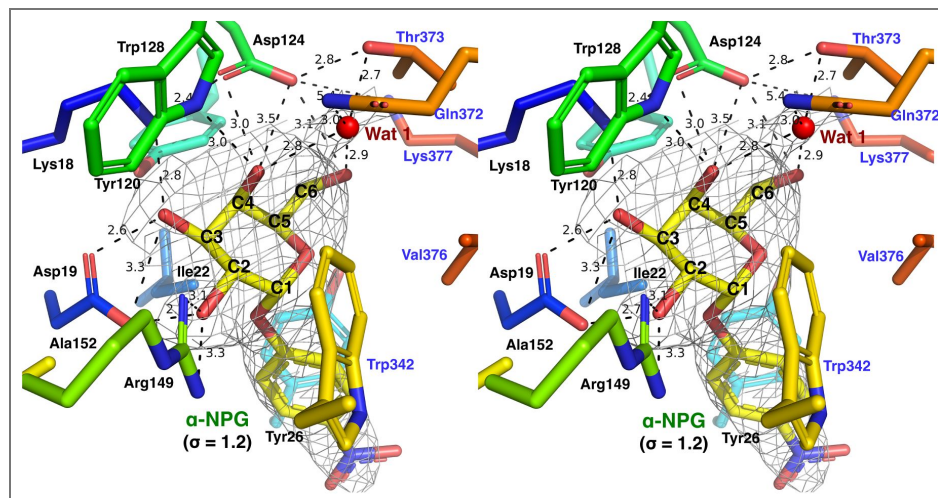


Fig. 3. α -NPG binding.

The cross-eye stereo view of α -NPG binding. The side chains forming the binding pocket formed from N- and C-terminal domains were shown in stick representation, colored according to corresponding hosting helices in rainbow, and labeled in black and blue, respectively. Isomesh map of the α -NPG (in yellow) and Wat 1 (in red) were contoured at a level of $\sigma = 1.2$. Dashed lines indicate distances within hydrogen-bonding or salt-bridge interactions (Å).



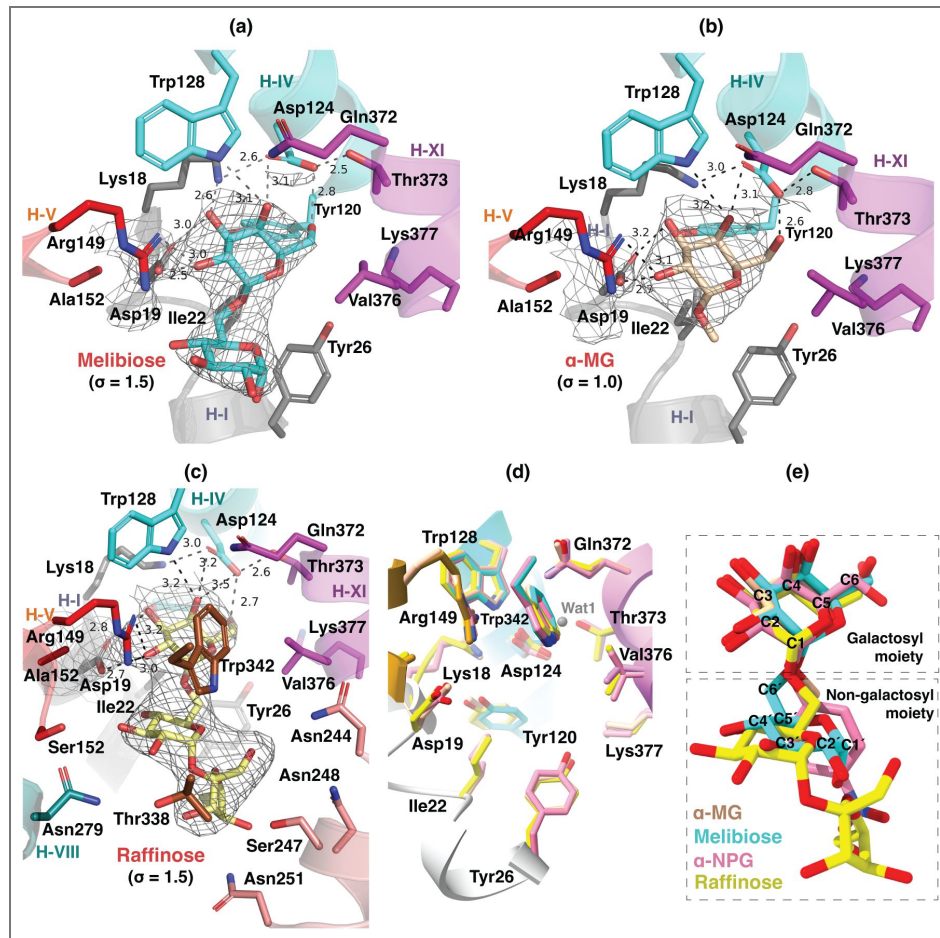


Fig. 4. Binding of melibiose, α -MG, and raffinose.

All residues within a 5-Å distance to the bound substrates were shown in sticks. Dashed lines, the distances within hydrogen-bonding and salt-bridge interactions (Å). **(a)** Melibiose binding. **(b)** α -MG binding. **(c)** Raffinose binding. **(d)** Residues in the sugar-binding pockets from the alignment of all four structures. **(e)** Substrates from the alignment of all four structures. Carbon positions on the galactosyl (C1-6) and glucosyl moiety (C1'-6') were labeled on the melibiose molecule. Trp342 was removed for clarity in panels a, b, and d. Isomesh maps for each sugar and Asp19 were contoured at levels of $\sigma = 1.5$ for melibiose and raffinose or $\sigma = 1.0$ for α -MG.

Structural dynamics measured by HDX-MS

To determine the structural dynamics of MelB_{St} and the influence of melibiose and/or Na⁺, the differential HDX-MS experiments were employed between MelB_{St} alone and MelB_{St} with melibiose and/or Na⁺ (28, 32, 33) (Table S2). MelB_{St} residues 2-470 out of 476 were covered, achieving a coverage of over 87.47% and 86.62% from 149 or 152 overlapping deuterium-labeled peptides (Tables S2-3; Fig. S4). There are 59, 63, or 59 non-covered residues for the datasets of the apo vs. melibiose-bound, Na⁺-bound, and melibiose- and Na⁺-bound states, respectively. Most are at the transmembrane helices IV, IX, X, and XI, and some cytoplasmic loops. Labeled peptides cover all periplasmic loops.

Six regions with greater deuteriation at the apo state

The peptide coverage-based deuteriation per residue plot and the relative deuterium uptake per peptide were generated from mean values from all three time points of each peptide with six duplicates (Fig. 5a,b; Table S3). The data indicate that, aside from the N- and C-terminal tails, which exhibit high HDX, six regions showed greater deuterium uptake in the apo state. As highlighted in red, two were in the N-terminal helices (I and V) and four were in the C-terminal loops (loop₆₋₇, loop₈₋₉, loop₉₋₁₀, and loop₁₀₋₁₁), which are the dynamic regions of MelB_{St} in the apo state. Notably, all the ICH1-3 were in the dynamic area. Comparing with helices I and V carrying the sugar-binding residues, helices II and IV—housing both the Na⁺-binding residues (Asp55, Asn58, and Asp59) and part of the sugar-binding residues (Asp124 and Trp128)—exhibited significantly lower deuteriation levels.

The effects of substrate(s) on MelB_{St} dynamics

The differential deuterium labeling (ΔD), which was calculated based on deuterium uptake in the absence (apo state) or presence of specific ligand(s) (Holo state), was shown as the residual plot of each peptide at three labeling time points and the sum of all (Fig. 6a). The dashed lines indicate the global thresholds calculated for each dataset. A hybrid significance analysis was used to determine the significance: $\Delta D >$ the global threshold of each dataset and $P < 0.05$ (34). More than 50% of residues (237, 264, or 257 positions) exhibited either insignificant ΔD values ($D_{\text{Mel}} - \text{Apo} < |0.186|$, $\Delta D_{(\text{Na}^+)} - \text{Apo} < |0.224|$, or $\Delta D_{\text{Na}^+(\text{Mel})} - \text{Apo} < |0.175|$), or $P > 0.05$, respectively (Table S2). Melibiose binding induced a wide range of effects on HDX, including a few protections (less deuterium uptake in the Holo state) and more deprotections (greater deuterium uptake in the Holo state). In contrast, Na⁺ alone or with melibiose primarily caused deprotections. The results supported the previous conclusion determined by the thermal denaturation study detected by circular dichroism spectroscopy (30); i.e., the melibiose-, or Na⁺-bound MelB_{St} was more stable than the apo state, and when MelB_{St} bound with both, it was the best. All deprotected peptides with significance were labeled individually (Fig. 6a). A peptide with at least one significant ΔD (meeting both criteria) from any time points was mapped onto the melibiose-bound structure (Fig. 6b-d; Fig. S4b).

Consensus effects were observed in all three labeling conditions (Fig. 6a-d), and peptides with significant effects were clustered in several regions, which largely overlapped with the dynamic regions obtained at the apo state (Fig. 5). Four out of six dynamic regions showed protection. Nearly full-length helices I and V, and regions 213-226 covering ICH1, showed protection by melibiose and/or Na⁺. The peptides covering 364-374 at the loop₁₀₋₁₁ and the starting part of helix XI, which contains the sugar-binding residues Gln372 and Thr373, were protected by Na⁺ or Na⁺ with melibiose. Another two dynamic regions exhibited diverse effects. The cytoplasmic positions 292-298 covering the loop₈₋₉ and the starting helix IX, containing a conformation-important residue Arg295, were deprotected by melibiose but protected by melibiose and Na⁺. The neighboring peptide 284-291, corresponding to helix VIII and loop₈₋₉ covering ICH2 at the cytoplasmic gating area, was also deprotected at the 30-sec time point but protected at the 3000-sec time point by melibiose and Na⁺ (Figs. 6d). The positions 317-326 at helix IX-loop₉₋₁₀-helix X showed deprotections by melibiose but were protected by Na⁺. Five out of six loops, except for loop₅₋₆ (Fig. 6a-b, pink and red), were deprotected by melibiose. In addition, the lid ICH3 was

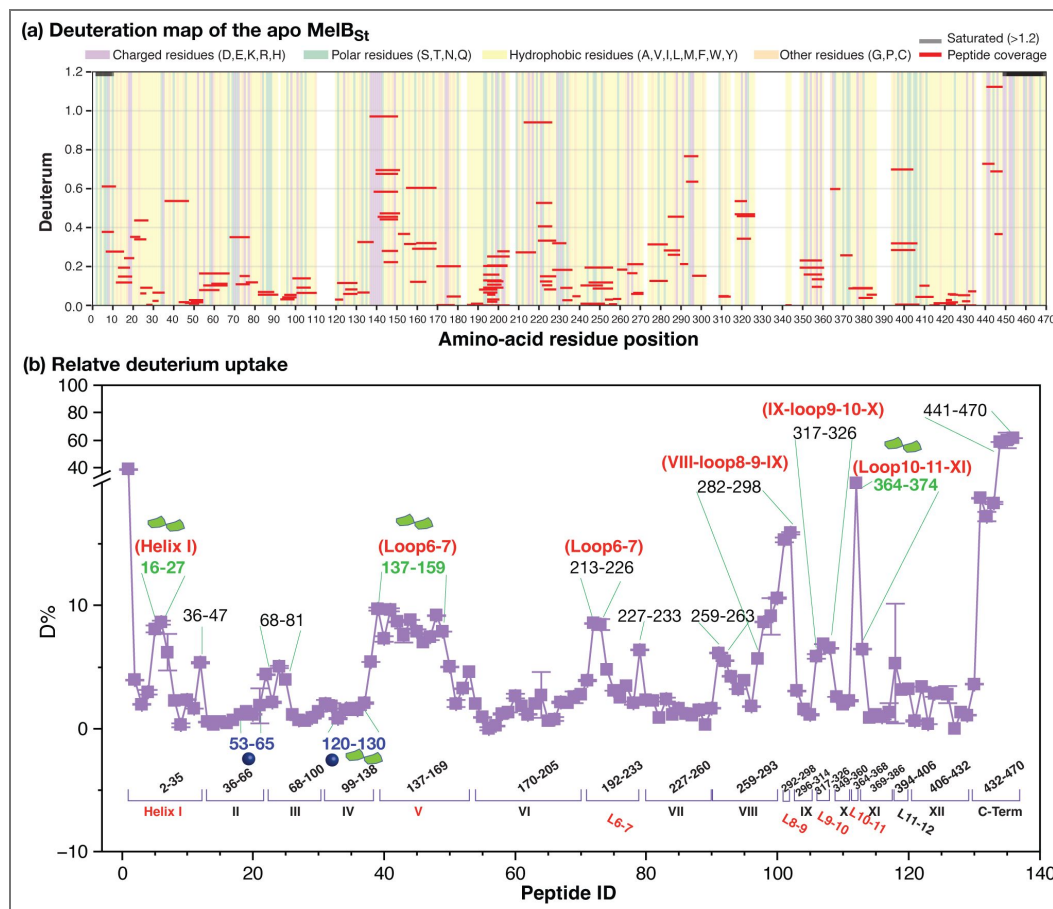


Fig. 5. Deuterium uptake of the apo MelB_{St}.

HDX experiments on WT MelB_{St} were conducted as described in Methods. All values were from the mean of six measurements at the apo state. **(a)** Deuteration map of the apo MelB_{St}. Mean deuteration levels of MelB_{St} peptides at the apo state averaged first across timepoints (30s, 300s, 3000s), then across two replicates, were presented against amino-acid residue sequence. The values at both N- and C-terminal regions were greater than 1.2 showed as black bars. The chemical properties of the peptide-covered residue were indicated by background shading; the white background indicated the non-covered positions. **(b)** Relative deuterium uptake per peptide plot. The corresponding transmembrane helices and a few loops were marked. Notably, due to the nature of overlapping peptides, the indicated amino-acid position is not sequential. Peptides with greater deuterium uptake were labeled. Peptides covering sugar- and cation-binding sites were colored in green and blue, respectively, and the dynamic regions with greater uptakes were colored in red.

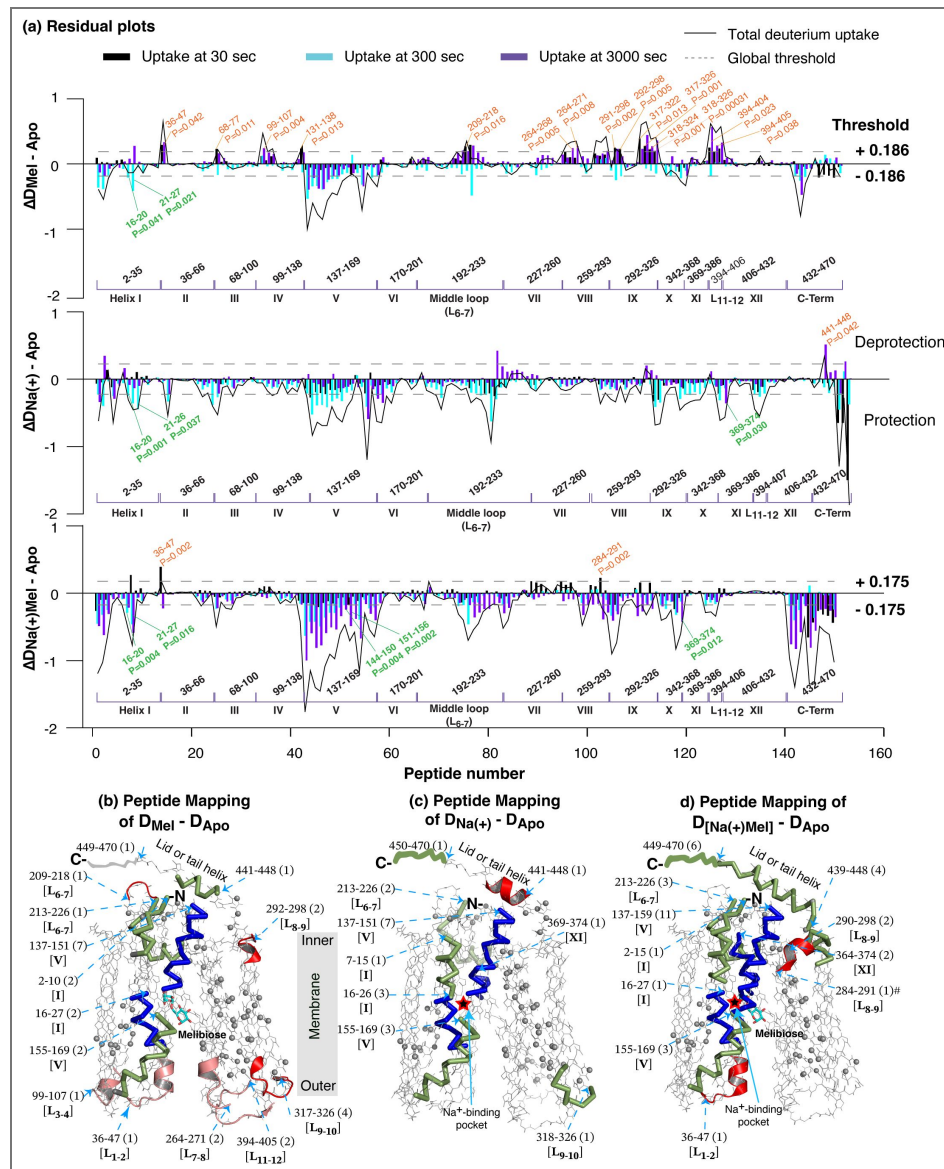


Fig. 6. Residual plots and structural mapping.

HDX experiments on WT MelB_{St} in the apo or holo states (with melibiose, Na⁺, or Na⁺ plus melibiose) were conducted as described in Methods. **(a)** Residual plots ($D_{Holo} - A_{po}$). Differential deuterium uptakes (ΔD) of each time point and the total uptake calculated from paired conditions for position 2-470 were plotted against the peptide number. Black, cyan, and purple bars, the deuterium uptake at 30, 300, and 3000 sec, respectively; dark gray curve, total uptake from all three time points. Deprotection, $\Delta D_{Holo} - A_{po} > 0$; protection, $\Delta D_{Holo} - A_{po} < 0$. Each sample was analyzed in triplicate. Dashed lines indicate the levels of the global threshold values calculated from each dataset, as labeled. The protein residue positions corresponding to the overlapping peptides were marked, and the covered transmembrane helices were labeled in Roman numerals. All deprotected peptides with statistical insignificance ($\Delta D > \text{threshold}$ and $P < 0.05$) in each dataset were labeled. **(b-d)** Peptide mapping on the crystal structure of the melibiose-bound inward-facing conformation for $D_{Mel} - A_{po}$, $D_{Na(+)} - A_{po}$, and $D_{Na(+)/Mel} - A_{po}$, respectively. The red star symbol indicated the location of the Na⁺-binding pocket, the drawing lines represented the disordered MelB_{St} C-terminal tail, and a gray bar showed the membrane region of MelB_{St}. The peptides of ΔD values with statistical significance ($\Delta D > |\text{threshold}|$ and $P < 0.05$) at any timepoint are highlighted either in ribbon representation for protection (colored in blue for peptides covering sugar-binding pocket and in green for all other regions) or in cartoon representation for deprotection (colored pink for data from 3000 sec and red for data from 30 sec). Non-covered residues from each dataset are shown as gray spheres at the Ca position and listed in Table S3, and peptides with statistically insignificant differences (either $\Delta D < |\text{threshold}|$ or $P > 0.05$) are illustrated in backbone representation in gray. Peptide positions are marked by their starting residue; the number of overlapping peptides is indicated in round brackets, and the location is shown in square brackets.

protected by melibiose or melibiose with Na^+ , with a single result of deprotection by Na^+ . The following sections will focus on the sugar- and cation-binding pockets as well as the structural elements critical for conformational transitions. All peptides with ΔD values greater than the threshold value were selected, and their deuterium uptake time course plots were presented in the Supplemental Figure S5.

A flexible sugar-binding pocket with a rigid cation-binding pocket

This HDX study covered most positions for the binding pockets for sugar and Na^+ (Table S4). Ten representative peptides were highlighted by uptake time course plots with structural mapping (Fig. 7 [7](#)). Three peptides 53-62 (helix II) and 120-123 & 121-130 (helix IV), which cover all Na^+ -binding residues (Asp55, Asn58, Asp59, and Thr121), exhibited lower deuteriation levels at the apo state (Figs. 5 [5](#), 7 [7](#), black open squares) with no significant effect by melibiose and/or Na^+ (Fig. 7 [7](#), blue filled squares). Others in helices II and IV also exhibited similar behavior (Fig. 5 [5](#); Table S3), indicating that both helices, including the Na^+ -binding residues, are conformationally rigid.

As described, helices I and V are the major flexible transmembrane helices (Fig. 5 [5](#)), which cover the six sugar-binding residues (Lys18, Asp20, Ile22, Tyr26, Arg149, and Ala152). Peptides 16-20 and 21-27 (helix I) were significantly protected by melibiose, Na^+ that binds remotely, or both (Fig. 7 [7](#)). Arg149 (helix V) was well covered by several overlapping peptides that consistently demonstrated greater protections by melibiose and/or Na^+ (Fig. 6 [6](#)-7 [7](#)). The shorter peptide 144-150 (helix V) showed protection by melibiose or Na^+ alone, which is statistically significant; however, the magnitude of the change is subtle. In the presence of both, the protection became significant. The Ala152-carrying peptide showed significant protection by melibiose and Na^+ , and peptide 368-273, which covers the sugar-binding residues Gln372 and Thr373 (helix XI), also exhibited significant protection by Na^+ alone or in combination with melibiose. Peptides carrying other sugar-binding residues Asp124, Try128, and Trp342 (helix IV) and Val376/Lys377 (XI) showed poor deuteriation and no significant effects by either melibiose, Na^+ or both. Overall, the results indicated that the sugar-binding residues in proximity to the cation-binding pocket are rigid with no significant effect by either substrate; in contrary, peptides carrying the sugar-binding residues far from the cation-binding site are dynamic, and their flexibility were significantly inhibited by sugar binding itself, by Na^+ alone, especially by the binding of both, which support that the melibiose affinity is correlated with the conformational flexibility and Na^+ can increase the sugar binding.

Dynamics of the cytoplasmic gating salt-bridge network and modulations by substrates

The cytoplasmic gating salt-bridge network between the two domains, involving nine charged residues (Fig. 8a, c [c](#)), was most covered. Residues Lys138, Arg141, and Glu142 at loop₄₋₅-helix V are three critical positions from the N-terminal domain. Overlapping peptides that covered this region consistently showed greater deuteriation levels at the apo state and greater protections under all three conditions, as represented by the peptide 137-150 (Figs. 5 [5](#)-7 [7](#)). Notably, the sugar-binding residue Arg149 at the gate area of helix V is within this most dynamic area of the transmembrane domain of MelB_{St} and the same dynamics group. Another gating residue, Arg295 (helix IX), which forms multiple interactions in this network, including two water molecules (Fig. S2), was covered by peptide 292-298, which showed a higher deuteriation level and was deprotected by melibiose but protected by melibiose and Na^+ . Peptide 364-368 at Loop₁₀₋₁₁ and the connected region of helix XI, which contains Glu365 at the other side of this salt-bridge network, also showed high deuterium uptake and protection by melibiose with Na^+ . On the contrary, Asp341, Asp354, and Glu357 at the rigid helix X, which are located in the center of this salt-bridge network, showed poor deuteriation under all conditions. The results showed that gating residues surrounding the rigid helix X in this broadly spanned cytoplasmic salt-bridge network are highly dynamic, and sensitive to the binding of melibiose, Na^+ , and melibiose with Na^+ .

As shown by the aligned outward- and inward-facing conformations and the membrane topology (Fig. 8b-c [c](#)), two gating regions at ICH2 and helix I-loop₁₋₂-helix II, covered by C-terminal peptide 284-291 and N-terminal peptide 36-47 at the cytoplasmic or periplasmic side, respectively, were

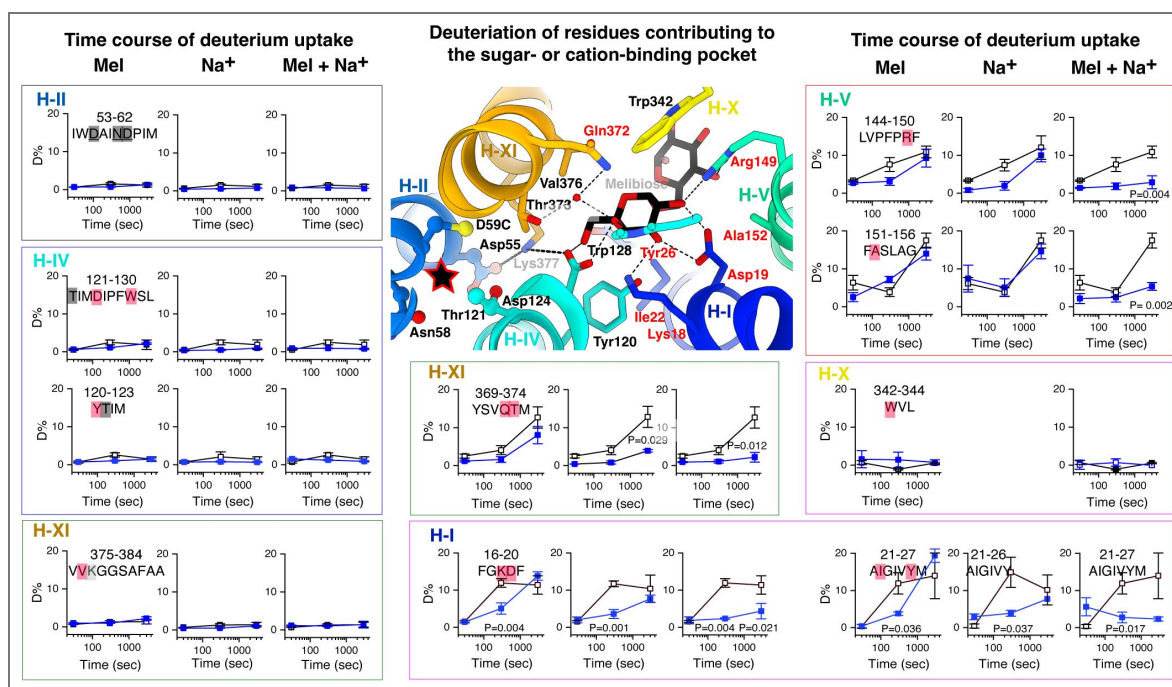


Fig. 7. Allosteric effects on substrate-binding sites.

Deuterium uptake time course of representative peptides covering the sugar- and cation-binding pockets. The percentage of deuterium uptake measured in the absence (empty black square) or presence of melibiose (Mel), Na⁺, or Mel & Na⁺ (filled blue square) was plotted against labeling times of 0, 30, 300, and 3000 sec. The peptide sequences were shown with residues at the cation- or sugar-binding pocket highlighted in black or red, respectively. Lys377 between the two binding pockets was highlighted in gray. P values are provided for each time point where the ΔD value exceeds the threshold. On the melibiose-bound structure, residues participating in the sugar binding and cation binding were shown in stick. Red star, the cation-binding pocket; red text labels: residues showing greater HDX with significant substrate effects; black text labels: residues showing poor HDX with no significant substrate effects.

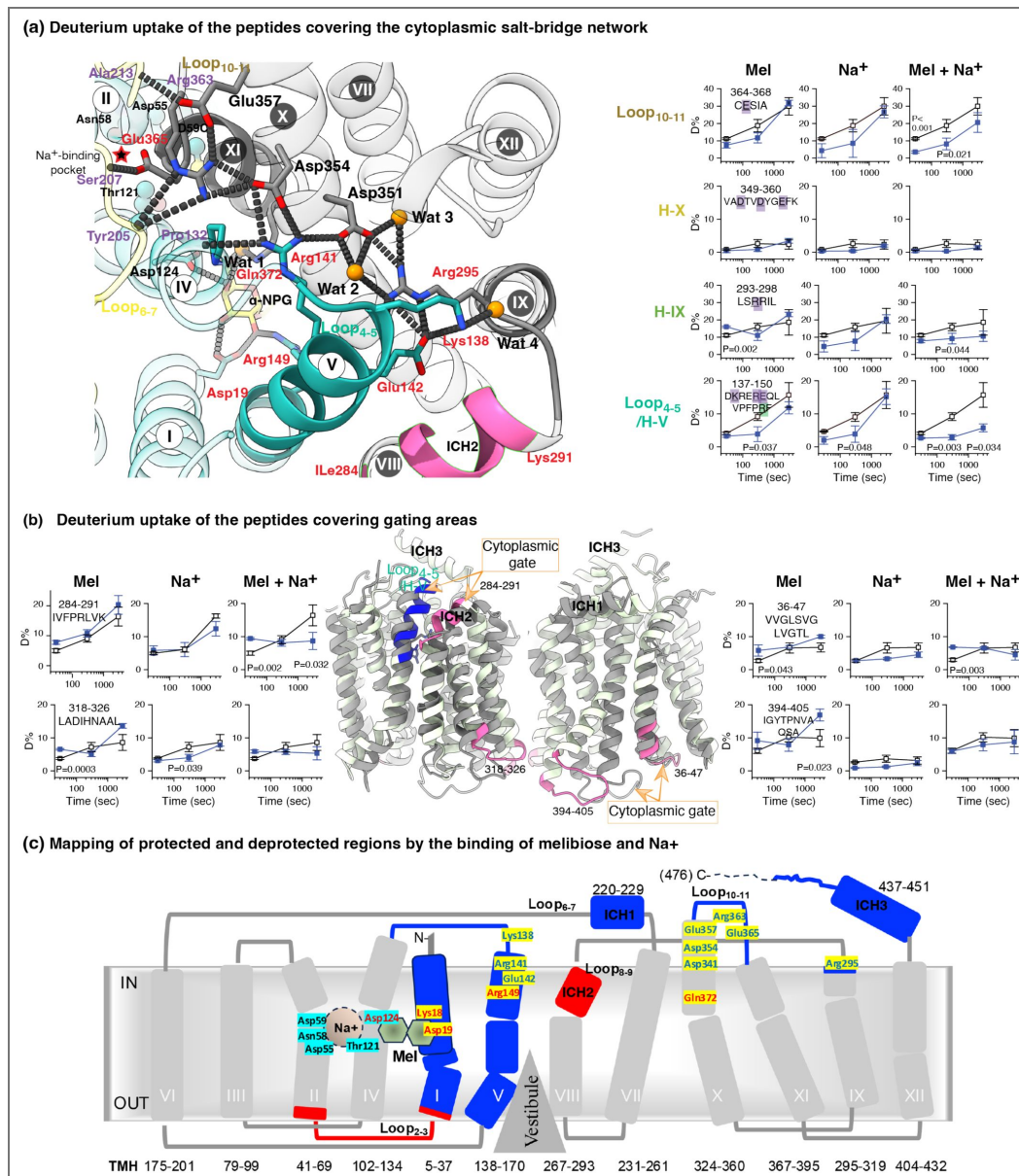


Fig. 8. Allosteric effects on gating salt-bridge network.

(a) Deuterium uptake time course of representative peptides covering the cytoplasmic gating salt-bridge network. The percentage of deuterium uptake measured in the absence (empty black square) or presence of melibiose (Mel), Na⁺, or Mel & Na⁺ (filled blue square) was plotted against labeling times of 0, 30, 300, and 3000 sec. P values are provided for each time point where the ΔD value exceeds the threshold. The peptide sequences were shown with residues at the salt-bridge network or sugar-binding pocket highlighted in purple or red, respectively. On the α -NPG bound structure, the gating salt-bridge network (Lys138, Arg141, and Glu142 (Loop₄₋₅/IV), Arg295 (IX), Asp351, Asp354, and Glu357 (X), and Arg363 and Asp365 (Loop₁₀₋₁₂) and their polar contacts with the backbone of other positions were shown in dashed lines. The N- and C-terminal domains were colored in light cyan and gray, respectively. The residues at the cation binding pocket were shown in ball and stick, as also indicated by the red star. Wat, water. Three sugar-binding residues, Asp19, Arg149, and Gln272, were shown in stick. Arg363 is in the list of uncovered positions. The region covering both the cytoplasmic gating salt-bridge network and sugar-binding residue Arg149 was highlighted in solid color, and the peptide 284-291 at helix III-ICH2 was also highlighted in pink. **(b)** Deprotection at loops. Deuterium uptake time course of four peptides at loops, mainly at the gating area, was presented and also mapped on the outward-facing structure, which was overlaid with the inward-facing structure [PDB 8T60]. The cytoplasmic and periplasmic gates were indicated. The peptides with deprotection by substrate were colored in pink. **(c)** HDX and ligand effects mapping on an outward-facing topology model of MelB₅. Melibiose- and Na⁺-binding residues were labeled in red or black, respectively. Residues showing ligand-induced protection and deprotection were colored in blue or red, respectively.

deprotected by melibiose or together with Na^+ . In addition, another gating region at loop₁₁₋₁₂ covered by the C-terminal peptide 394-405 and the peptide 318-326 at helix XI-loop₉₋₁₀-helix X was also deprotected by melibiose. Most deprotected areas are at the periplasmic loops.

Molecular dynamics (MD) simulations

Three systems, including the apo, melibiose-bound, and melibiose- and Na^+ -bound states of a generated WT MelB_{St}, were constructed by embedding in a POPE:POPG (7:2) lipid bilayer. For each system setup, five independent replicas of MD simulations were carried out, each for ~400 ns, and the total sampling time for each system was ~2 μs .

Water 1 occupancy

To analyze the Wat 1 modeled in the higher resolution structure with α -NPG bound, a distance-based criterion for defining the average water occupancy in the sugar-binding site across all trajectories of each system as described in Methods. Results showed that Wat-1 exhibited nearly full occupancy when melibiose was present, regardless of whether Na^+ was bound at the cation-binding site (Table S6), which supported the crystal structure observation.

Side-chain flexibility

The side-chain heavy-atom root-mean-square fluctuation (RMSF) over five replicas of trajectories was analyzed for the apo and MelB_{St} with melibiose and Na^+ bound states (Fig. 9 [↗](#)). The results showed that the side chains in the cation-binding site and nearby sugar-binding positions reduced fluctuations upon the binding of substrates. The residues Asp19, Tyr26, Arg149, and Gln372 at helices I, V, or XI, respectively, were relatively flexible, and their conformational freedoms were significantly reduced by the binding of melibiose and Na^+ . The data about the sidechain flexibility are consistent with the peptide-based HDX results.

Discussion

The binding recognition and affinity of MelB_{St} have been further refined

Dehydration of sugar molecules is expected for binding; however, full or partial dehydration in MelB is unknown. The 2.60 Å resolution α -NPG-bound structure showed a partially dehydrated sugar at the binding site. The bound water connected the OH-4 and OH-6 on the galactopyranosyl ring with Thr373 and Gln372 at helix XI, and it was also surrounded by Asp124 at helix IV. Previous Cys-scanning mutagenesis has shown that the T373C mutant retained most activities. Still, a Cys residue on the Gln372 position significantly decreased the transport initial rate, accumulation, and melibiose fermentation, with little effect on the protein expression ([35](#)), which supported the role of Wat-1 in the binding. Notably, the orientation of OH-4 is crucial for distinguishing between galactose and glucose. Previously, the OH-4 has been shown at hydrogen-bonding distances from the carboxyl group of Asp124 and the indole group of Trp128 at Helix IV ([20](#), [22](#)), and the identification of Wat-1 in this study added the polar residues Gln372 and Thr373 on helix XI to stabilize the critical OH-4 (Fig. 3 [↗](#)). Thus, these interactions define the specificity of galactosyl molecules and Wat-1 is likely part of sugar binding. Notably, the OH-4 and OH-6 and Wat 1 are in close proximity to the Na^+ -binding residue Thr121 and the cation-binding important residue Lys377, implying that the Wat-1 is between the two specificity-determining pockets. As shown previously, the OH-3 and OH-2 at the opposite edge of the galactopyranosyl ring form multiple hydrogen-bonding interactions with the charged residues Asp19 and Arg149 at helices I and V, which can be assigned to play a crucial role in stabilizing the recognition of OH-4 and enhancing the binding affinity.

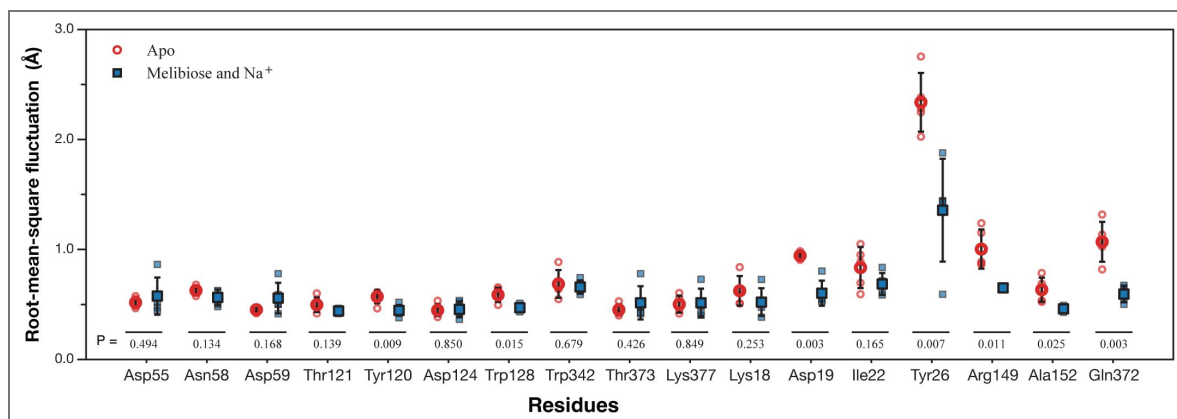


Fig. 9. Side-chain flexibility analyzed from MD trajectories.

For each residue, the side-chain RMSF values in each of the five replicas for the apo and melibiose- and Na⁺-bound states were plotted. The mean of the RMSF values in each state is represented as red circle (apo state) or blue square (melibiose- and Na⁺-bound state). For each residue, unpaired t test of the RMSF mean values between the apo and the melibiose- and Na⁺-bound states were performed and the p-values were presented.

Dynamics of MelB_{St} at different regions

HDX-MS is a powerful technique that simultaneously discloses the dynamic information in various areas of a protein. The qualitative method overcomes the drawbacks of most site-specific labeling-based techniques; however, it only provides the conformational dynamic information on the equilibrium of all ensembles. In the current study, we determined the deuterium uptake rates of the full-length MelB_{St} in the absence or presence of melibiose and/or Na⁺, identified major dynamic regions (Fig. 5 [5](#)), as well as substrate-induced effects on the substrate-binding sites and structural elements critical for conformational transition, including the cytoplasmic salt-bridge network and the gating areas (Figs. 6 [6](#)-8 [8](#)).

Our studies of HDX compensated with molecular dynamics simulations on side-chain fluctuations showed that the Na⁺-binding pocket and the sugar-binding residues near the Na⁺-binding site at helix IV are conformationally rigid since they exhibited low deuteration level for both unbound and bound states (Figs. 5 [5](#)-7 [7](#), 9 [9](#); Tables S3-4), which are consistent to the previous HDX-MS results of conformation transition (21). Those rigidities are likely derived from their hosting helices (II and IV). In addition, the Na⁺-binding affinity, different from sugar binding, exhibited little difference between the inward- and outward-facing states (21). The conformational freedom of the cation-binding site and most sugar recognition positions was likely restricted in most states to facilitate sugar binding.

The sugar-binding residues, located away from the Na⁺-binding site, are dynamic, exhibiting higher deuterium levels and side-chain fluctuations. Most of those positions are within the six dynamic regions identified from the apo state, including Lys18, Asp19, Ile22, Tyr26, Arg149, Ala152, Gln272, and Thr373 at helices I, V, and XI. Those regions were protected to varying extents by melibiose or Na⁺, and the protection was significantly greater when bound to both substrates (Figs. 5 [5](#)-7 [7](#)). Even their conformational flexibility was restrained by melibiose and Na⁺, but the deuterium uptakes at those regions were still greater than those of peptides covering the cation-binding site (Fig. 7 [7](#)). Notably, substrate-induced protection was also detected in other areas of the hosting helices I and V, as further stressed below.

Arg149, a sugar-binding residue in the major dynamic gating area

Abundant peptides covering Arg149 consistently showed protection by melibiose or Na⁺, especially when both were present (Figs. 5 [5](#)-8 [8](#)). Structurally, Arg149 is near to a stretch of charged residues as shown by this peptide 137-DK¹³⁷RER¹⁴¹E¹⁴²QLVPFPRF-150 (Figs. 5 [5](#)-7 [7](#)), where the Lys138, Arg141, and Glu142 have been determined to play critical roles in the functionally crucial cytoplasmic salt-bridge network (Fig. 8 [8](#)). The dynamics of Arg149 are likely modulated by this charged network and belongs to a same dynamics group, which provided the structural basis for coupling between the sugar-binding affinity and protein dynamics. At the inward-open structure, this network is deformed, Arg149, as part of this gate, exhibits a large displacement due to an otherwise stereo collision with helix X at the outward-facing state (Fig. S6). The sugar-binding affinity at this inward-facing state was reduced by ~30-fold due to the broken binding pocket and the displacement of Arg149 and Gln372. Functionally, the single-site R149C mutant inhibited but did not eliminate the melibiose transport (18), and single-Cys149 (35) also fermented melibiose with intact cells. The neighboring P148L mutant decreased transport $V_{\max}$ with a better K_m (36). All data showed that the dynamics of this charged network and sugar-binding affinity are interconnected.

This cytoplasmic gating salt-bridge network has a rigid center dictated by helix XI and dynamic surroundings from helices V, IX, XI, loops 4-5, 8-9, 6-7, 10-11, and those dynamic positions were significantly inhibited by melibiose, Na⁺, especially the binding of both. Notably, Nb725 binding to the inward-facing conformation also inhibited the dynamics of the similar regions as shown by peptides 137–150, 284–289, and 364–368 (21). The similar HDX profiles from all testing conditions suggested the protein motions followed a similar pathway of the conformational transition. Thus, the dynamics of Arg149 are linked to motion and conformational transitions, as well as sugar-binding affinity. Connectively, it is postulated that Na⁺ binding between helices II and IV stabilizes

the dynamics of the gating salt-bridge network and the flexibility of Arg149 and Asp19, which are located more than 10 Å away, thus switching the primary substrate-binding site to the higher-affinity state allosterically.

The binding of sugar further stabilizes this network and promotes Na⁺ binding by preventing Na⁺ from leaving the cation-binding pocket, thus increasing Na⁺-binding affinity (18, 26). Therefore, the cooperative binding of sugar and Na⁺ is part of the substrate-induced conformational allostery, and all functions together.

There are a few regions with substrate-induced deprotections. Two gating areas were deprotected by melibiose and Na⁺, including the cytoplasmic ICH2 and Loop₁₋₂, which suggested the structural arrangements. Notably, ICH2 is linked with Arg295 at one end of the salt-bridge network, and the dynamics of this area could influence the stability of the gating network, which might trigger the separation and conformational transition to the inward-open state. Thus, the increased dynamics of loop₁₋₂ and Loop₈₋₉ at both gating areas by melibiose, especially with Na⁺, can be interpreted as the tendency to form a transition-competent conformation with closing at the periplasmic gate and opening at the cytoplasmic gate. A region at loop₆₋₇ responding to the ligand binding identified by the atomic force microscopy was not observed in this HDX study (37).

The symporter H⁺-coupled lactose permease LacY and xylose permease Xyle also show protonation dependence of sugar-binding affinity (32, 38, 39), with separate cation- and sugar-binding sites (5, 40–43). Both transporters favored inward-facing conformation and substrate binding induced outward-facing conformation, as indicated by the extensive HDX analysis with Xyle (32, 44) and a bunch of biophysical measurements of LacY (45). Notably, several dynamic regions in MelB_{St} were also identified in Xyle; both proteins showed binding-induced structural changes in their loop₁₋₂ and loop₈₋₉ (44). In MelB_{St}, while no quantified information on the ensembles, all experimental results and molecular simulations on the minimum free-energy landscape for sugar translocation suggest that the apo MelB favors the outward-facing conformation, and the outward-facing conformation is further preferred when bound with melibiose and Na⁺ (20, 21, 26).

Summary

Our studies on structure, HDX, and MD simulations allow us to conclude that the sugar-binding affinity of MelB_{St} is coordinated with protein structural motions and conformational transition between inward- and outward-facing states. Na⁺ binding restrains the dynamics of remote sugar-binding residues via stabilizing the dynamic cytoplasmic salt-bridge network, thereby increasing sugar-binding affinity allosterically. The conclusion provides insightful knowledge to understand cooperative binding and symport mechanisms.

Experimental Procedures

Reagents

Melibiose was purchased from Acros Organics (ThermoFisher Scientific). [³H]Raffinose and label-free raffinose were purchased from American Radiolabeled Chemicals (ARC), Inc and Research Products International (RPI) Corp., respectively. Raffinose (RPI Chemicals), α-methyl galactoside (α-MG) and *p*-Nitrophenyl α-D-galactoside (α-NPG) were purchased from Sigma-Aldrich. Detergents undecyl-β-D-maltopyranoside (UDM) and dodecyl-β-D-maltopyranoside (DDM) were purchased from Anatrace. *E. coli* lipids (Extract Polar, 100600) were purchased from Avanti Polar Lipids, Inc. All other materials were reagent grade and obtained from commercial sources.

Strains and plasmids

E. coli DW2 cells (*melA*⁺*B*⁺, *lacZ*⁺*Y*) (46) were used for protein expression and functional studies. The expression plasmids pK95/ΔAH/WT MelB_{St}/CHis₁₀ (15) and pK95/ΔAH/D59CMelB_{St}/CHis₁₀ (18) were used for constitutive expression.

MelB_{St} protein expression and purification

Cell growth for the large-scale production of WT MelB_{St} or D59C MelB_{St} was carried out in *E. coli* DW2 cells (18, 47). Briefly, MelB_{St} purification by cobalt-affinity chromatography (Talon Superflow Metal Affinity Resin, Takara) after extraction by 1.5% UDM. MelB_{St} protein was eluted with 250 mM imidazole in a buffer containing 50 mM NaPi, pH 7.5, 200 mM NaCl, 0.035% UDM, and 10% glycerol, and further dialyzed to change the buffer conditions accordingly.

Protein concentration assay

The Micro BCA Protein Assay (Pierce Biotechnology, Inc.) was used to determine the protein concentration.

α-NPG transport

MelB_{St}-mediated α-NPG downhill transport was detected by the release of the intracellular α-NPG hydrolytic product, *p*-nitrophenol, from *E. coli* DW2 cells expressing high-turnover number α-galactosidase and recombinantly expressed MelB_{St} (Ottina, 1980). *E. coli* DW2 cells carrying the constitutive expression plasmid for the WT MelB_{St} or D59C MelB_{St} uniporter mutant in LB media containing 100 mg/L ampicillin were grown in a 37 °C shaker as described (15). The overnight cultures were diluted 5-fold into fresh LB broth and 100 mg/L ampicillin, and then shaken at 30 °C for 4 h. The expression of α-galactosidase was induced by adding 5 mM melibiose for 1 hour before harvesting the cells for the transport assay. The melibiose-induced cells were washed with 100 mM KP_i, pH 7.5, three times to remove the remaining melibiose and Na⁺, and adjusted to A₄₂₀ of 10 (~0.7 mg proteins/ml) in the assay solution of 100 mM KP_i, pH 7.5, 10 mM MgSO₄, and 1 mM DTT in the absence or presence of 20 mM NaCl or LiCl. The cells under each condition were equilibrated at a 30 °C incubator for 10 minutes before the α-NPG transport assay, which was initiated by mixing 0.5 mM label-free α-NPG and incubating for 60 min. A 100-μL cell aliquot at the given time point of 0, 1, 5, 10, 20, 30, 60 min was quenched with 900-μL 0.33 M Na₂CO₃, followed by a centrifugation at 10,000 g for 5 min to collect the clarified supernatant for absorbance measurements. The extracellular *p*-nitrophenol released from the intracellular hydrolytic product of the translocated α-NPG, which was supplied in the extracellular environment, was measured at 405 nm by a UV spectrometer. The concentration of *p*-nitrophenol is estimated based on its molar extinction coefficient of 18000 M⁻¹cm⁻¹.

[³H]Raffinose transport

The raffinose active transport was carried out by [³H]raffinose uptake with DW2 cells expressing MelB_{St} without the induction of α-galactosidase as described for the melibiose transport assay (15). Briefly, the cells expressing the WT MelB_{St} or D59C MelB_{St} uniporter mutant, or without a plasmid, were mixed with 2-μL of 25 mM [³H]raffinose (specific activity 10 mCi/mmol) to 50 μL of cells (final raffinose concentration at 1 mM in the absence or presence of 50 mM NaCl or LiCl, and the transport reaction was quenched at the given time points and followed by a fast filtration.

Isothermal titration calorimetry

All ITC ligand-binding assays were performed with the TA Instruments (Nano-ITC device) as described (19), which yields the exothermic binding as a positive peak. The MelB_{St} was dialyzed overnight with assay buffer containing 20 mM Tris-HCl (pH 7.5), 100 mM NaCl, 10% glycerol, and 0.035% UDM. The ligands are prepared by dissolving in the same batch of dialysis buffer for buffer matching. In a typical experiment, the titrand (MelB_{St}) placed in the ITC Sample Cell was titrated with the specified titrant raffinose or α-methyl galactoside (placed in the Syringe) in the assay buffer by an incremental injection of 2 or 2.5-μL aliquots at an interval of 250 or 300 sec at a constant stirring rate of 250 rpm (nano-ITC). MelB_{St} protein samples were buffer-matched to the assay buffer by dialysis. The normalized heat changes were subtracted from the heat of dilution elicited by the last few injections, where no further binding occurred, and the corrected heat

changes were plotted against the mole ratio of the titrant to the titrand. The values for the binding association constant (K_a) were obtained by fitting the data using the one-site independent-binding model included in the NanoAnalyze software (version 3.7.5). The dissociation constant (K_d) = $1/K_a$.

Crystallization, native diffraction data collection, and processing

The D59C MelB_{St} dialyzed overnight against the sugar-free dialysis buffer (20 mM Tris-HCl, pH 7.5, 100 mM NaCl, 0.035% UDM, and 10% glycerol), concentrated with Vivaspin column at 50 kDa cutoff, and stored at -80 °C. A phospholipid stock solution of 20 mM was prepared by dissolving the *E. coli* Extract Polar (Avanti, 100600) with a dialysis buffer containing 0.01% DDM. The protein sample was diluted to a final concentration of 10 mg/ml with the same sugar-free dialysis buffer, supplemented with phospholipids at 3.6 mM and 30 mM of melibiose or α -MG, 40 mM raffinose, or 6 mM of α -NPG in DMSO solution. Crystallization trials were conducted using the hanging-drop vapor-diffusion method at 23°C by mixing 2 μ L of protein with 2 μ L of reservoir solution. Crystals from D59C MelB_{St} protein with the melibiose, α -MG, or α -NPG appeared against a reservoir consisting of 100 mM Tris-HCl, pH 8.5, 100 mM NaCl₂, 50 mM CaCl₂, and 32-35% PEG 400. For the raffinose-containing sample, the crystals were collected from 100 mM Tris-HCl, pH 8.5, 50 mM CaCl₂, 50 mM BaCl₂, and 32.5% PEG 400. All crystals were frozen in liquid nitrogen within 2 weeks and tested for X-ray diffraction at the Lawrence Berkeley National Laboratory ALS beamlines 5.0.1 (for the melibiose, α -MG, and α -NPG-containing complex datasets) or 5.0.2 (for the raffinose-containing datasets) using the remote data collection method.

ALS auto-processing XDS or DIALS programs output files were further reduction by AIMLESS in the ccp4i2 program for the structure solution (48). The statistics in data collection are described in Table S1.

Structure determination

The structure determination was performed by the Molecular Replacement method using the α -NPG-bound D59C MelB_{St} mutant structure [PDB ID 7L17] as the search template, followed by rounds of manual building and refinement to resolutions of 2.60 Å, 3.05, 3.45, or 3.68 Å for structure with α -NPG, melibiose, raffinose, or α -MG, respectively, in Phenix (49). The model building and refinement were performed in Phenix and Coot (50), respectively. The structures were modeled from positions 2 to 453 or 455, respectively, without gaps, and the missing side chains due to density disorder, as well as the Ramachandran assessment, are listed in Table S5.

Sugar docking and modeling

One strong positive density, with varying sizes and shapes, was observed in the difference maps of each of the four structures. The size and shape matched the sugars that co-crystallized, and the docked sugar molecules fitted well with the densities. The sugar refinement restraints were generated from SMILES using the ELBOW program in Phenix (49). To the 2.60 Å α -NPG-bound map, five water molecules were modeled. In addition, a PEG molecule (ligand ID 1PE) was also modeled to a strong positive density with a sausage shape aligning with helix IX. To the melibiose-bound structure, two water molecules were added.

Hydrogen-deuterium exchange coupled to mass spectrometry (HDX-MS)

An in-solution HDX-MS experiment was performed to study the substrate-induced structural dynamics of MelB_{St}. As described (21), the labeling, quenching, lipid removal, and online digestion were achieved using a fully automated manner using an HDx3 extended parallel system (LEAP Technologies, Morrisville, NC) (51, 52). MelB_{St} was prepared at 50.0 μ M in a Na⁺-free buffer (25 mM Tris-HCl, pH 7.5, 150 mM choline chloride, 10% glycerol, 0.035% UDM in H₂O), either in the absence of a substrate (apo) or in the presence of 100 mM melibiose, 100 mM NaCl, or both. The hydrogen/deuterium exchange reaction as described previously (21). Briefly, aliquots of 4 μ L of each sample were diluted 10-fold into the labeling buffer (25 mM Tris-HCl, pH 7.5, 50 mM choline chloride, 10% glycerol, 0.035% UDM in D₂O), without or with 100 mM of melibiose, Na⁺, or both.

Labeled samples were incubated in D₂O buffer at 20 °C for multiple time points (30 sec, 300 sec, and 3000 sec) in triplicate, and non-deuterated controls were prepared similarly, except that H₂O buffer was used in the labeling step.

At each designated time point, reaction was quenched by adding an equal volume of ice-cold quench buffer (6 M urea, 100 mM citric acid, pH, 2.3 in H₂O) for 180 seconds at 0 °C and immediately subjected to a lipid filtration module integrated on the LEAP PAL system. After incubation of a 60 sec with ZrO₂ particles, the LEAP X-Press then compressed the filter assembly to separate proteins from the ZrO₂ particles-bound phospholipids and detergents. The filtered protein sample was injected into a cool box for online digestion and separation.

LC/MS bottom-up HDX was performed using a Thermo Scientific™ Ultimate™ 3000 UHPLC system and Thermo Scientific™ Orbitrap Eclipse™ Tribrid™ mass spectrometer. Samples were digested with a Nepenthesin-2 (Affipro, Czech Republic) column at 8 °C and then trapped in a 1.0 mm x 5.0 mm, 5.0 μm trap cartridge for desalting over 180 sec. The resulting peptides were then separated on a Thermo Scientific™ Hypersil Gold™, 50 x1 mm, 1.9 μm, C18 column with a gradient of 10 % to 40 % gradient (A: water, 0.1 % formic acid; B: acetonitrile, 0.1 % formic acid) for 15 minutes at a flow rate of 40 μL/min. A pepsin wash was added in between runs to minimize the carryover.

A nonspecific digested peptide database has been created for MelB_{St} with a separate MS/MS measurement of non-deuterated samples as described (21). Digested peptides from undeuterated MelB_{St} protein were identified on the orbitrap mass spectrometer using the same LC gradient as the HDX-MS experiment. Using the Thermo BioPharma Finder software (v 5.1), MS2 spectra were matched to the MelB_{St} sequence with fixed modifications.

A total of 146 or 150 peptide assignments (with confident HDX data across all labeling times) were confirmed for MelB_{St} samples, resulting in 86-87% sequence coverage. The MS data were processed using the Sierra Analytics HDEaminer software with the MelB_{St} peptide database. Following the automated HDX-MS analysis, manual curation was performed. Upon the completion of the data review, a single charge state with high-quality spectra for all replicates across all HDX labeling times was chosen to represent HDX for each peptide. Differential HDX data were tested for statistical significance using the hybrid significance testing criteria method with an in-house MATLAB script, where the HDX differences at different protein states were calculated ($\Delta D = D_{\text{Holo}} - D_{\text{Apo}}$). Mean HDX differences from the three replicates were assigned as significant according to the hybrid criteria based on the pooled standard deviation and Welch's t-test with $P < 0.05$. The statistically significant differences observed at each residue ($\Delta D_{\text{Mel}} - \text{Apo} < |0.186|$, $\Delta D_{(\text{Na}^+)} - \text{Apo} < |0.224|$, or $\Delta D_{\text{Na}^+(\text{Mel})} - \text{Apo} < |0.175|$) were used to map HDX consensus effects based on overlapping peptides onto the structure models.

Statistics and reproducibility

All experiments were performed 2-4 times. The average values were presented in the table with standard errors. An unpaired t-test was used for statistical analysis. For the relative D%, the data were transferred to log values prior to the unpaired t-test.

Graphs

Pymol (3.1.5.1) (53) and UCSF ChimeraX (X) were used to generate all graphs. The program Origin 2024 was used to plot the ITC curves and transport data.

MD simulations

The crystal structure of the α-NPG-bound D59C MelB_{St} at a resolution of 2.6 Å was replaced with a melibiose molecule according to the melibiose-bound structure, and the D59C was mutated back to Asp. Both Asp59 and Asp55 residues were set in the deprotonated state. Three systems, including the apo, melibiose-bound, and melibiose- and Na⁺-bound states, were generated by embedding them in a POPE:POPG (7:2) lipid bilayer created using the CHARMM-GUI(54) web server. The lipid bilayer was capped with a 25 Å water box on each side, and ~0.15 M NaCl was added to neutralize the system charge and mimic the ionic strength of physiological conditions. The resulting system

has ~120,000 atoms, with a periodic boundary condition of $\sim 110 \times 110 \times 125$ (Å). The ff14SB(55), lipid17(56), and GLYCAM(57) force fields were employed to treat the protein, lipid, and melibiose, respectively. The TIP3P(58) water model was used for all water molecules.

After system assembly, energy minimization was carried out with harmonic restraints (1000 kJ/mol/Å²) on protein and lipid's heavy atoms for 40,000 steps. This was followed by 200 ps of equilibration in the constant NVT ensemble at 300 K and 1 ns of NPT equilibration with gradually decreased force constants in the harmonic restraints. Five independent replicas of production simulations in the constant NPT ensemble were initiated from different snapshots in the NVT trajectory. For each replica, a trajectory was propagated for ~400 ns at 300 K and 1 atm. The total sampling time for each system was ~2 μs. The temperature was controlled with a Langevin thermostat(59) with a friction coefficient of 1 ps⁻¹ and pressure with a Berendsen barostat(60) using anisotropic scaling with a relaxation time of 1 ps. Long-range electrostatics were treated with the Particle Mesh Ewald (PME) method(61) (tolerance 5×10^{-4}), and van der Waals interactions were truncated using a 12 Å cutoff distance. A 2 fs timestep was used throughout all simulations. The SHAKE algorithm(62) was employed to constrain the lengths of all hydrogen-containing covalent bonds. All MD simulations were performed with the AMBER24 software package.(63)

Water-1 occupancy identified in the α-NPG-bound crystal structure was evaluated by monitoring interactions of water molecules with four coordinating residues. For each frame, a water molecule was defined as located in the sugar-binding pocket if the distances from its oxygen (O) atom to a few neighboring residues and the melibiose molecule satisfied the following rules. First, four pairs of heavy atoms on the melibiose molecule and neighboring residues were defined: O5 on the galactosyl ring and NE2 on Gln372, O5 and OG1 on Thr373, O6 and NE2 on Gln372, as well as O6 and OG1 on Thr373. Second, for each pair of the above-defined heavy atoms, the distance between the water oxygen atom (O) and each of the two heavy atoms in the pair was calculated. Third, out of all four pairs, if at least in one of them both distances were within 4 Å and at least one distance was below 3.5 Å, the water molecule was defined as occupying the sugar-binding site. The water-1 occupancy in the sugar-binding site was then computed as the fraction of frames containing at least one occupying water molecule over the total number of frames in all replicas of the trajectories.

The side-chain heavy-atom root-mean-square fluctuation (RMSF) of all residues in the sugar- and cation-binding pockets were determined using CPPTRAJ.

Data availability

The x-ray diffraction datasets and models have been deposited to wwPDB under the accession codes 9OLD for the α-nitrophenyl galactoside-bound, 9OLI for the melibiose-bound, 9OLR for the α-methyl galactoside, as well as 9OLP for the raffinose-bound D59C MelB_{St}.

Acknowledgements

The authors thank Dr. William Mallard for creating Fig. 5a [\[link\]](#). The X-ray diffraction datasets were collected at ALS BL 5.01 or 5.0.2. This work was supported by the National Institutes of Health Grants R35GM153222 to L.G. and R35GM150780 to R.L.

Abbreviations

MelB_{St}: *Salmonella enterica* serovar Typhimurium melibiose permease
MFS: major facilitator superfamilies
ITC: isothermal titration calorimetry
α-NPG: *p*-nitrophenyl-α-D-galactopyranoside
Mel: melibiose
α-MG: methyl α-D-galactoside
Raff: raffinose
*K*_d: dissociation constant

HDX-MS: hydrogen-deuterium exchange coupled to mass spectrometry
D: deuterium
RMSF: root-mean-square fluctuation

Additional files

[Supplemental Table S1](#)

[Supplemental Table S2](#)

[Supplemental Table S3](#)

[Supplemental Table S4](#)

[Supplemental Table S5](#)

[Supplemental Table S6](#)

[Supplemental Figs. 1-6](#)

Additional information

Funding

Funder	Grant reference number	Author
Nation Institute (Type Media Center)	R35 GM153222	Lan Guan
Nation Institute (Type Media Center)	R35 GM150780	Ruibin Liang

Author ORCID iDs

Parameswaran Hariharan: <https://orcid.org/0000-0002-6020-1547>

Lan Guan: <https://orcid.org/0000-0002-2274-361X>

References

- Ferrada E.,** Superti-Furga G (2022) A structure and evolutionary-based classification of solute carriers. *iScience* **25**:105096
- Pao S. S.,** Paulsen I. T., Saier M. H. (1998) Major facilitator superfamily. *Microbiol Mol Biol Rev* **62**:1-32
- Cesar-Razquin A.,** Snijder B., Frappier-Brinton T., Isserlin R., Gyimesi G., Bai X., et al. (2015) A Call for Systematic Research on Solute Carriers. *Cell* **162**:478-487
- Lin L.,** Yee S. W., Kim R. B., Giacomini K. M (2015) SLC transporters as therapeutic targets: emerging opportunities. *Nat Rev Drug Discov* **14**:543-560
- Guan L.,** Kaback H. R (2006) Lessons from lactose permease. *Annu Rev Biophys Biomol Struct* **35**:67-91
- Yan N** (2015) Structural Biology of the Major Facilitator Superfamily Transporters. *Annu Rev Biophys* **44**:257-283
- Drew D.,** North R. A., Nagarathinam K., Tanabe M (2021) Structures and General Transport Mechanisms by the Major Facilitator Superfamily (MFS). *Chem Rev* **121**:5289-5335
- Guan L** (2023) The rapid developments of membrane protein structure biology over the last two decades. *BMC Biol* **21**:300
- Nguyen L. N.,** Ma D., Shui G., Wong P., Cazenave-Gassiot A., Zhang X., et al. (2014) Mfsd2a is a transporter for the essential omega-3 fatty acid docosahexaenoic acid. *Nature* **509**:503-506
- Cater R. J.,** Chua G. L., Erramilli S. K., Keener J. E., Choy B. C., Tokarz P., et al. (2021) Structural basis of omega-3 fatty acid transport across the blood-brain barrier. *Nature* **595**:315-319

11. Chua G. L., Tan B. C., Loke R. Y. J., He M., Chin C. F., Wong B. H., et al. (2023) Mfsd2a utilizes a flippase mechanism to mediate omega-3 fatty acid lysolipid transport. *Proc Natl Acad Sci U S A* **120**:e2215290120
12. Wilson T. H., Ding P. Z (2001) Sodium-substrate cotransport in bacteria. *Biochim Biophys Acta* **1505**:121-130
13. Maehrel C., Cordat E., Mus-Veteau I., Leblanc G (1998) Structural studies of the melibiose permease of *Escherichia coli* by fluorescence resonance energy transfer. I. Evidence for ion-induced conformational change. *J Biol Chem* **273**:33192-33197
14. Meyer-Lipp K., Sery N., Ganea C., Basquin C., Fendler K., Leblanc G (2006) The inner interhelix loop 4-5 of the melibiose permease from *Escherichia coli* takes part in conformational changes after sugar binding. *J Biol Chem* **281**:25882-25892
15. Guan L., Nurva S., Ankeshwarapu S. P (2011) Mechanism of melibiose/cation symport of the melibiose permease of *Salmonella typhimurium*. *J Biol Chem* **286**:6367-6374
16. Granell M., Leon X., Leblanc G., Padros E., Lorenz-Fonfria V. A (2010) Structural insights into the activation mechanism of melibiose permease by sodium binding. *Proc Natl Acad Sci U S A* **107**:22078-22083
17. Guan L (2018) Na(+)/Melibiose Membrane Transport Protein, MelB. In: Roberts G., Watts A., European Biophysical Societies, Roberts G, Watts A (Eds). *Encyclopedia of Biophysics* Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-35943-9_10082-1
18. Ethayathulla A. S., Yousef M. S., Amin A., Leblanc G., Kaback H. R., Guan L (2014) Structure-based mechanism for Na(+)/melibiose symport by MelB. *Nat Commun* **5**:3009
19. Hariharan P., Guan L (2017) Thermodynamic cooperativity of cosubstrate binding and cation selectivity of *Salmonella typhimurium* MelB. *J Gen Physiol* **149**:1029-1039
20. Guan L., Hariharan P (2021) X-ray crystallography reveals molecular recognition mechanism for sugar binding in a melibiose transporter MelB. *Commun Biol* **4**:931
21. Hariharan P., Shi Y., Katsube S., Willibal K., Burrows N. D., Mitchell P., et al. (2024) Mobile barrier mechanisms for Na(+)-coupled symport in an MFS sugar transporter. *eLife* **12**
22. Hariharan P., Bakhtiari A., Liang R., Guan L (2024) Distinct roles of the major binding residues in the cation-binding pocket of the melibiose transporter MelB. *J Biol Chem* 107427
23. Katsube S., Liang R., Amin A., Hariharan P., Guan L (2022) Molecular Basis for the Cation Selectivity of *Salmonella typhimurium* Melibiose Permease. *J Mol Biol* **434**:167598
24. Guan L., Jakkula S. V., Hodkoff A. A., Su Y (2012) Role of Gly117 in the cation/melibiose symport of MelB of *Salmonella typhimurium*. *Biochemistry* **51**:2950-2957
25. Hariharan P., Guan L (2014) Insights into the inhibitory mechanisms of the regulatory protein IIA(Glc) on melibiose permease activity. *J Biol Chem* **289**:33012-33019
26. Liang R., Guan L. (2024) Atomic-Level Free Energy Landscape Reveals Cooperative Symport Mechanism of Melibiose Transporter. *eLife*. <https://doi.org/10.7554/eLife.103421>
27. Hamuro Y (2024) Interpretation of Hydrogen/Deuterium Exchange Mass Spectrometry. *J Am Soc Mass Spectrom* **35**:819-828
28. Masson G. R., Burke J. E., Ahn N. G., Anand G. S., Borchers C., Brier S., et al. (2019) Recommendations for performing, interpreting and reporting hydrogen deuterium exchange mass spectrometry (HDX-MS) experiments. *Nat Methods* **16**:595-602
29. Wilson D. M., Wilson T. H (1987) Cation specificity for sugar substrates of the melibiose carrier in *Escherichia coli*. *Biochim Biophys Acta* **904**:191-200
30. Hariharan P., Guan L (2021) Cooperative binding ensures the obligatory melibiose/Na⁺ cotransport in MelB. *J Gen Physiol* **153**
31. Amin A., Ethayathulla A. S., Guan L (2014) Suppression of conformation-compromised mutants of *Salmonella enterica* serovar Typhimurium MelB. *J Bacteriol* **196**:3134-3139

32. Jia R., Martens C., Shekhar M., Pant S., Pellowe G. A., Lau A. M., et al. (2020) Hydrogen-deuterium exchange mass spectrometry captures distinct dynamics upon substrate and inhibitor binding to a transporter. *Nat Commun* **11**:6162
33. Zmyslowski A. M., Baxa M. C., Gagnon I. A., Sosnick T. R (2022) HDX-MS performed on BtuB in *E. coli* outer membranes delineates the luminal domain's allostery and unfolding upon B12 and TonB binding. *Proc Natl Acad Sci U S A* **119**:e2119436119
34. Hageman T. S., Weis D. D (2019) Reliable Identification of Significant Differences in Differential Hydrogen Exchange-Mass Spectrometry Measurements Using a Hybrid Significance Testing Approach. *Anal Chem* **91**:8008-8016
35. Markham K. J., Tikhonova E. B., Scarpa A. C., Hariharan P., Katsube S., Guan L (2021) Complete cysteine-scanning mutagenesis of the *Salmonella typhimurium* melibiose permease. *J Biol Chem* **297**:101090
36. Jakkula S. V., Guan L (2012) Reduced Na(+) affinity increases turnover of *Salmonella enterica* serovar Typhimurium MelB. *J Bacteriol* **194**:5538-5544
37. Blaimschein N., Hariharan P., Manioglou S., Guan L., Muller D. J (2023) Substrate-binding guides individual melibiose permeases MelB to structurally soften and to destabilize cytoplasmic middle-loop C3. *Structure* **31**:58-67.
38. Smirnova I. N., Kasho V., Kaback H. R (2008) Protonation and sugar binding to LacY. *Proc Natl Acad Sci U S A* **105**:8896-8901
39. Grytsyk N., Sugihara J., Kaback H. R., Hellwig P (2017) pKa of Glu325 in LacY. *Proc Natl Acad Sci U S A* **114**:1530-1535
40. Guan L., Mirza O., Verner G., Iwata S., Kaback H (2007) Structural determination of wild-type lactose permease. *Proceedings of the National Academy of Sciences of the United States of America* **104**:15294-15298
41. Mirza O., Guan L., Verner G., Iwata S., Kaback H. R (2006) Structural evidence for induced fit and a mechanism for sugar/H⁺ symport in LacY. *EMBO J* **25**:1177-1183
42. Kumar H., Kasho V., Smirnova I., Finer-Moore J. S., Kaback H. R., Stroud R. M (2014) Structure of sugar-bound LacY. *Proc Natl Acad Sci U S A* **111**:1784-1788
43. Sun L., Zeng X., Yan C., Sun X., Gong X., Rao Y., et al. (2012) Crystal structure of a bacterial homologue of glucose transporters GLUT1-4. *Nature* **490**:361-366
44. Jia R., Bradshaw R. T., Calvaresi V., Politis A (2023) Integrating Hydrogen Deuterium Exchange-Mass Spectrometry with Molecular Simulations Enables Quantification of the Conformational Populations of the Sugar Transporter XylE. *J Am Chem Soc* **145**:7768-7779
45. Smirnova I., Kasho V., Kaback H. R (2011) Lactose permease and the alternating access mechanism. *Biochemistry* **50**:9684-9693
46. Botfield M. C., Wilson T. H (1988) Mutations that simultaneously alter both sugar and cation specificity in the melibiose carrier of *Escherichia coli*. *J Biol Chem* **263**:12909-12915
47. Pourcher T., Leclercq S., Brandolin G., Leblanc G (1995) Melibiose permease of *Escherichia coli*: large scale purification and evidence that H(+), Na(+), and Li(+) sugar symport is catalyzed by a single polypeptide. *Biochemistry* **34**:4412-4420
48. Potterton L., Agirre J., Ballard C., Cowtan K., Dodson E., Evans P. R., et al. (2018) CCP4i2: the new graphical user interface to the CCP4 program suite. *Acta Crystallogr D Struct Biol* **74**:68-84
49. Liebschner D., Afonine P. V., Baker M. L., Bunkoczi G., Chen V. B., Croll T. I., et al. (2019) Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. *Acta Crystallogr D Struct Biol* **75**:861-877
50. Casanal A., Lohkamp B., Emsley P (2020) Current developments in Coot for macromolecular model building of Electron Cryo-microscopy and Crystallographic Data. *Protein Sci* **29**:1069-1078
51. Hamuro Y., Coales S. J., Southern M. R., Nemeth-Cawley J. F., Stranz D. D., Griffin P. R (2003) Rapid analysis of protein structure and dynamics by hydrogen/deuterium exchange mass spectrometry. *J Biomol Tech* **14**:171-182

52. Hamuro Y., Coales S. J (2018) Optimization of Feasibility Stage for Hydrogen/Deuterium Exchange Mass Spectrometry. *J Am Soc Mass Spectrom* **29**:623-629
53. Schrodinger L. (2013) The PyMOL Molecular Graphics System.
54. Wu E. L., Cheng X., Jo S., Rui H., Song K. C., Dávila Contreras E. M., et al. (2014) CHARMM GUI membrane builder toward realistic biological membrane simulations. *J Comput Chem*
55. Maier J. A., Martinez C., Kasavajhala K., Wickstrom L., Hauser K. E., Simmerling C (2015) ff14SB: improving the accuracy of protein side chain and backbone parameters from ff99SB. *Journal of chemical theory and computation* **11**:3696-3713
56. Skjevik Å. A., Madej B. D., Dickson C. J., Lin C., Teigen K., Walker R. C., et al. (2016) Simulation of lipid bilayer self-assembly using all-atom lipid force fields. *Phys Chem Chem Phys* **18**:10573-10584
57. (no date) GLYCAM Web Complex Carbohydrate Research Center. <http://glycam.org>
58. Jorgensen W. L., Chandrasekhar J., Madura J. D., Impey R. W., Klein M. L (1983) Comparison of simple potential functions for simulating liquid water. *J Chem Phys* **79**:926-935
59. Sindhikara D. J., Kim S., Voter A. F., Roitberg A. E (2009) Bad seeds sprout perilous dynamics: Stochastic thermostat induced trajectory synchronization in biomolecules. *Journal of Chemical Theory and Computation* **5**:1624-1631
60. Uberuaga B. P., Anghel M., Voter A. F (2004) Synchronization of trajectories in canonical molecular-dynamics simulations: Observation, explanation, and exploitation. *J Chem Phys* **120**:6363-6374
61. Darden T., York D., Pedersen L (1993) Particle mesh Ewald: An N log (N) method for Ewald sums in large systems. *J Chem Phys* **98**:10089-10089
62. Ryckaert J.-P., Ciccotti G., Berendsen H. J (1977) Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes. *J Comput Phys* **23**:327-341
63. Case D. A., Aktulga H. M., Belfon K., Ben-Shalom I. Y., Berryman J. T., Brozell S. R., et al. (2023) *Amber 2023* San Francisco: University of California.

Peer reviews

Reviewer #1 (Public review):

While the structure of the melibiose permease in both outward and inward-facing forms has been solved previously, there remains unanswered questions regarding its mechanism. Hariharan et al set out to address this with further crystallographic studies complemented with ITC and hydrogen deuterium exchange (HDX) mass spectrometry. They first report 4 different crystal structures of galactose derivatives to explore molecular recognition showing that the galactose moiety itself is the main source of specificity. Interestingly, they observe a water-mediated hydrogen bonding interaction with the protein and suggest that this water molecule may be important in binding.

The results from the crystallography appear sensible, though the resolution of the data is low with only the structure with NPG better than 3Å. Support for the conclusion of the water molecule in the binding site, as interpreted from the density, is given by MD studies.

The HDX also appears to be well done and is explained reasonably well in the revision.

<https://doi.org/10.7554/eLife.108335.2.sa2>

Reviewer #3 (Public review):

Summary:

The melibiose permease from *Salmonella enterica* serovar Typhimurium (MelBSt) is a member of the Major Facilitator Superfamily (MFS). It catalyzes the symport of a

galactopyranoside with Na⁺, H⁺, or Li⁺, and serves as a prototype model system for investigating cation-coupled transport mechanisms. In cation-coupled symporters, a coupling cation typically moves down its electrochemical gradient to drive the uphill transport of a primary substrate; however, the precise role and molecular contribution of the cation in substrate binding and translocation remain unclear. In a prior study, the authors showed that the binding affinity for melibiose is increased in the presence of Na⁺ by about 8-fold, but the molecular basis for the cooperative mechanism remains unclear. The objective of this study was to better understand the allosteric coupling between the Na⁺ and melibiose binding sites. To verify the sugar-recognition specific determinants, the authors solved the outward-facing crystal structures of a uniport mutant D59C with four sugar ligands containing different numbers of monosaccharide units (α-NPG, melibiose, raffinose, or α-MG). The structure with α-NPG bound has improved resolution (2.7 Å) compared to a previously published structure and to those with other sugars. These structures show that the specificity is clearly directed toward the galactosyl moiety. However, the increased affinity for α-NPG involves its hydrophobic phenyl group, positioned at 4 Å-distance from the phenyl group of Tyr26 forms a strong stacking interaction. Moreover, a water molecule bound to OH-4 in the structure with α-NPG was proposed to contribute to the sugar recognition and appears on the pathway between the two specificity-determining pockets. Next, the authors analyzed by hydrogen-to-deuterium exchange coupled to mass spectrometry (HDX-MS) the changes in structural dynamics of the transporter induced by melibiose, Na⁺, or both. The data support the conclusion that the binding of the coupling cation at a remote location stabilizes the sugar-binding residues to switch to a higher-affinity state. Therefore, the coupling cation in this symporter was proposed to be an allosteric activator.

Strengths:

- (1) The manuscript is generally well written.
- (2) This study builds on the authors' accumulated knowledge of the melibiose permease and integrates structural and HDX-MS analyses to better understand the communication between the sodium ion and sugar binding sites. A high sequence coverage was obtained for the HDX-MS data (86-87%), which is high for a membrane protein.

The revised manuscript shows clear improvement, and the authors have addressed my concerns in a satisfactory manner. Of note, I noticed two mistakes that should be corrected:

- page 11. Unless I am mistaken, the sentence "In contrast, Na⁺ alone or with melibiose primarily caused deprotections" should be corrected with "protections". The authors may wish to verify this sentence and also the previous one in the main text.

- Figure 8 displays two cytoplasmic gates (one of them should be periplasmic)

<https://doi.org/10.7554/eLife.108335.2.sa1>

Author response:

The following is the authors' response to the original reviews

eLife Assessment

This manuscript presents useful insights into the molecular basis underlying the positive cooperativity between the co-transported substrates (galactoside sugar and sodium ion) in the melibiose transporter MelB. Building on years of previous studies, this work improves on the resolution of previously published structures and reports the presence of a water molecule in the sugar binding site that would appear to be key for its recognition, introduces further structures bound to different substrates, and utilizes HDX-MS to further understand the positive cooperativity between sugar and the co-

transported sodium cation. Although the experimental work is solid, the presentation of the data lacks clarity, and in particular, the HDX-MS data interpretation requires further explanation in both methodology and discussion, as well as a clearer description of the new insight that is obtained in relation to previous studies. The work will be of interest to biologists and biochemists working on cation-coupled symporters, which mediate the transport of a wide range of solutes across cell membranes.

We express our gratitude to the associate editor, review editor, and reviewers for their favorable evaluation of this manuscript, as well as their constructive comments and encouragement. Their feedback has been integrated to fortify the evidence, refine the data analysis, and elevate the presentation of the results, thereby enhancing the overall quality and clarity of the manuscript.

A brief summary of the modifications in this revision:

(a) We performed four new experiments: 1) intact cell [^3H]raffinose transport assay; 2) intact cell *p*-nitrophenol detection to demonstrate α -NPG transport; 3) ITC binding assay for the D59C mutant; and 4) molecular dynamics to simulate the water-1 in sugar-binding site and the dynamics of side chains in the Na^+ - and melibiose-binding pockets. All data consistently support the conclusion drawn in this article.

(b) We have added a new figure to show the apo state dynamics (the new Fig. 5a,b) and annotated the amino acid residue positions and marked positions in sugar- or Na^+ -binding pockets.

(c) As suggested by reviewer-3, we have moved the individual mapping of ligand effects on HDX data to the main figure, combined with the residual plots, and marked the amino-acid residue positions.

(d) We have added more deuterium uptake plots to cover all residues in the sugar- or Na^+ -binding pockets in the current figure 7 (previously figure 6).

(e) We have added a new figure 8 showing the positions at the well-studied cytoplasmic gating salt-bridge network and other loops likely important for conformational changes, along with a membrane topology marked with the HDX data. We have added a new figure 9 from MD simulations.

Reviewer #1:

While the structure of the melibiose permease in both outward and inward-facing forms has been solved previously, there remain unanswered questions regarding its mechanism. Hariharan et al set out to address this with further crystallographic studies complemented with ITC and hydrogen-deuterium exchange (HDX) mass spectrometry.

(1) They first report 4 different crystal structures of galactose derivatives to explore molecular recognition, showing that the galactose moiety itself is the main source of specificity. Interestingly, they observe a water-mediated hydrogen bonding interaction with the protein and suggest that this water molecule may be important in binding.

We thank you for understanding what we've presented in this manuscript.

(2) The results from the crystallography appear sensible, though the resolution of the data is low, with only the structure with NPG better than 3Å. However, it is a bit difficult to understand what novel information is being brought out here and what is known about the ligands. For instance, are these molecules transported by the protein or do they just bind? They measure the affinity by ITC, but draw very few conclusions about how the affinity correlates with the binding modes. Can the protein transport the trisaccharide raffinose?

The four structures with bound sugars of different sizes were used to identify the binding motif on both the primary substrate (sugar) and the transporter (MelB_{ST}). Although the resolutions of the structures complexed with melibiose, raffinose, or α-MG are relatively low, the size and shape of the densities at each structure are consistent with the corresponding sugar molecules, which provide valuable data for confirming the pose of the bound sugar proposed previously. In this revision, we further refine the α-NPG-bound structure to 2.60 Å. The identified water-1 in this study further confirms the orientation of C4-OH. Notably, this transporter does not recognize or transport glucosides in which the orientation of the C4-OH at the glucopyranosyl ring is opposite. To verify the water in the sugar-binding site, we initiated a new collaborative study using MD simulations. Results showed that Wat-1 exhibited nearly full occupancy when melibiose was present, regardless of whether Na⁺ was bound at the cation-binding site.

As detailed in the Summary, we added two additional sets of transport assays and confirmed that raffinose and α-NPG are transportable substrates of MelB_{ST}. For α-NPG transport, we measured the end products of the process—enzyme hydrolysis and membrane diffusion of p-nitrophenol released from intracellular α-NPG.

As a bonus, based on the WT-like downhill α-NPG transport activity by the D59C uniporter mutant that failed in active transport against a sugar concentration gradient, we further emphasized that the sugar translocation pathway is isolated from the cation-binding site. The new data strongly support the allosteric effects of cation binding on sugar-binding affinity. Thank you for this helpful suggestion.

A meaningful analysis of ITC data heavily depends on the quality of the data. My laboratory has extensive experience with ITC and has gained rich, insightful mechanistic knowledge of MelB_{ST}. Because of the low affinity in raffinose and α-MG, unfortunately, no further information can be convincingly obtained. Therefore, we did not dissect the enthalpic and entropic contributions but focused on the K_d value and binding stoichiometry.

(3) The HDX also appears to be well done; however, in the manuscript as written, it is difficult to understand how this relates to the overall mechanism of the protein and the conformational changes that the protein undergoes.

We are sorry for not presenting our data clearly in the initial submission. In this revised manuscript, we have made numerous improvements, as described in the Summary. These enhancements in the HDX data analysis provided new mechanistic insights into the allosteric effects, leading us to conclude that protein dynamics and conformational transitions are coupled with sugar-binding affinity. Na⁺ binding restricts protein conformational flexibility, thereby increasing sugar-binding affinity. The HDX study revealed that the major dynamic region includes a sugar-binding residue, Arg149, which also plays a gating role. Structurally, this dual-function residue undergoes significant displacement during the sugar-affinity-coupled conformational transition, thereby coupling the sugar binding and structural dynamics.

Reviewer #2:

This manuscript from Hariharan, Shi, Viner, and Guan presents x-ray crystallographic structures of membrane protein MelB and HDX-MS analysis of ligand-induced dynamics. This work improves on the resolution of previously published structures, introduces further sugar-bound structures, and utilises HDX to explore in further depth the previously observed positive cooperativity to cotransported cation Na⁺. The work presented here builds on years of previous study and adds substantial new details into how Na⁺ binding facilitates melibiose binding and deepens the fundamental understanding of the molecular basis underlying the symport mechanism of cation-coupled transporters. However, the presentation of the data lacks clarity, and in

particular, the HDX-MS data interpretation requires further explanation in both methodology and discussion.

We appreciate this reviewer's time in reading our previous articles related to this manuscript.

Comments on Crystallography and biochemical work:

(1) It is not clear what Figure 2 is comparing. The text suggests this figure is a comparison of the lower resolution structure to the structure presented in this work; however, the figure legend does not mention which is which, and both images include a modelled water molecule that was not assigned due to poor resolution previously, as stated by the authors, in the previously generated structure. This figure should be more clearly explained.

This figure is a stereo view of a density map created in cross-eye style. In this revision, we changed this figure to Fig. 3 and showed only the density for sugar and water-1.

(2) It is slightly unclear what the ITC measurements add to this current manuscript. The authors comment that raffinose exhibiting poor binding affinity despite having more sugar units is surprising, but it is not surprising to me. No additional interactions can be mapped to these units on their structure, and while it fits into the substrate binding cavity, the extra bulk of additional sugar units is likely to reduce affinity. In fact, from their listed ITC measurements, this appears to be the trend. Additionally, the D59C mutant utilised here in structural determination is deficient in sodium/cation binding. The reported allostery of sodium-sugar binding will likely influence the sugar binding motif as represented by these structures. This is clearly represented by the authors' own ITC work. The ITC included in this work was carried out on the WT protein in the presence of Na⁺. The authors could benefit from clarifying how this work fits with the structural work or carrying out ITC with the D59C mutant, or additionally, in the absence of sodium.

Thank this reviewer for your helpful suggestions. We have performed the suggested ITC measurements with the D59C mutant. The purpose of the ITC experiments was to demonstrate that MelB_{St} can bind raffinose and α-MG to support the crystal structures.

Comments on HDX-MS work:

While the use of HDX-MS to deepen the understanding of ligand allostery is an elegant use of the technique, this reviewer advises the authors to refer to the Masson et al. (2019) recommendations for the HDX-MS article (<https://doi.org/10.1038/s41592-019-0459-y>) on how to best present this data. For example:

All authors value this reviewer's comments and suggestions, which have been included in this revision.

(1) The Methodology includes a lipid removal step. Based on other included methods, I assumed that the HDX-MS was being carried out in detergent-solubilised protein samples. I therefore do not see the need for a lipid removal step that is usually included for bilayer reconstituted samples. I note that this methodology is the same as previously used for MelB. It should be clarified why this step was included, if it was in fact used, aka, further details on the sample preparation should be included.

Yes, a lipid/detergent removal step was included in this study and previous ones, and this information was clearly described in the Methods.

(2) A summary of HDX conditions and results should be given as recommended, including the mean peptide length and average redundancy per state alongside other included information such as reaction temperature, sequence coverage, etc., as prepared for previous publications from the authors, i.e., Hariharan et al., 2024.

We have updated the Table S2 and addressed the reviewer's request for the details of HDX experiments.

(3) Uptake plots per peptide for the HDX-MS data should be included as supporting information outside of the few examples given in Figure 6.

We have prepared and presented deuterium uptake time-course plots for any peptides with $\Delta D > \text{threshold}$ in Fig. S5a-c.

*(4) A reference should be given to the hybrid significance testing method utilised. Additionally, as stated by Hageman and Weis (2019) (doi:10.1021/acs.analchem.9b01325), the use of $P < 0.05$ greatly increases the likelihood of false positive ΔD identifications. While the authors include multiple levels of significance, what they refer to as high and lower significant results, this reviewer understands that working with dynamic transporters can lead to increased data variation; a statement of why certain statistical criteria were chosen should be included, and possibly accompanied by volcano plots. The legend of Figure 6 should include what P value is meant by * and ** rather than statistically significant and highly statistically significant.*

We appreciate this comment and have cited the suggested article on the hybrid significance method. We fully acknowledge that using a cutoff of $P < 0.05$ can increase the likelihood of false-positive identifications. By applying multiple levels of statistical testing, we determined that $P < 0.05$ is an appropriate threshold for this study. The threshold values were presented in the residual plots and explained in the text. For the previous Fig. 6 (renamed Fig. S4b in the current version), we have reported the P value. *, < 0.05 ; **, < 0.01 . (The text for 0.01 was not visible in the previous version. Sorry for the confusion.)

(5) Line 316 states a significant difference in seen in dynamics, how is significance measured here? There is no S.D. given in Table S4. Can the authors further comment on the potential involvement in solvent accessibility and buried helices that might influence the overall dynamics outside of their role in sugar vs sodium binding? An expected low rate of exchange suggests that dynamics are likely influenced by solvent accessibility or peptide hydrophobicity. The increased dynamics at peptides covering the Na binding site on overall more dynamic helices suggests that there is no difference between the dynamics of each site.

The current Table S3 (combined from previous Tables S3 and S4 as suggested) was prepared to provide an overall view of the dynamic regions with SD values provided. For other questions, if we understand correctly, this reviewer asked us to comment on the effects of solvent accessibility or hydrophobic regions on the overall dynamics outside the binding residues of the peptides that cover them. Since HDX rates are influenced by two linked factors: solvent accessibility and hydrogen-bonding interactions that reflect structural dynamics, poor solvent accessibility in buried regions should result in low deuterium uptakes. The peptides in our dataset that include the Na^+ -binding site showed lower HDX, likely due to limited solvent accessibility and lower structural stability. It is unclear what this reviewer meant by "increased dynamics at peptides covering the Na binding site on overall more dynamic helices." We did not observe increased dynamics in peptides covering the Na^+ -binding site; instead, all Na^+ -binding residues and nearby sugar-binding residues have lower degrees of deuteration.

(6) Previously stated HDX-MS results of MelB (Hariharan et al., 2024) state that the transmembrane helices are less dynamic than polypeptide termini and loops with similar distributions across all transmembrane bundles. The previous data was obtained in the presence of sodium. Does this remove the difference in dynamics in the sugar-binding helices and the cation-binding helices? Including this comparison would support the

statement that the sodium-bound MelB is more stable than the Apo state, along with the lack of deprotection observed in the differential analysis.

Thanks for this suggestion. The previous datasets were collected in the presence of Na^+ . In the current study, we also have two Na^+ -containing datasets. Both showed similar results: the multiple overlapping peptides covering the sugar-binding residues on helices I and V have higher HDX rates than those peptides covering the Na^+ -binding residues, even when Na^+ was present.

(7) Have the authors considered carrying out an HDX-MS comparison between the WT and the D59C mutant? This may provide some further information on the WT structure (particularly a comparison with sugar-bound). This could be tied into a nice discussion of their structural data.

Thank you for this suggestion. Comparing HDX-MS between the WT and the D59C mutant is certainly interesting, especially with the increasing amount of structural, biochemical, and biophysical data now available for this mutant. However, due to limited resources, we might consider it later.

(8) Have the authors considered utilising Li^+ to infer how cation selectivity impacts the allostery? Do they expect similar stabilisation of a higher-affinity sugar binding state with all cations?

We have shown that Li^+ also works positively with melibiose. Li^+ binds to MelB_{St} with a higher affinity than Na^+ and modifies MelB_{St} differently. It is important to study this thoroughly and separately. To answer the second question, H^+ is a weak coupling cation with little effect on melibiose binding. Since its pKa is around 6.5, only a small population of MelB_{St} is protonated at pH 7.5. The order of sugar-binding cooperativity is highest with Na^+ , then Li^+ , and finally H^+ .

(9) MD of MelB suggests all transmembrane helices are reorientated during substrate translocation, yet substrate and cotransporter ligand binding only significantly impacts a small number of helices. Can the authors comment on the ensemble of states expected from each HDX experiment? The data presented here instead shows overall stabilisation of the transporter. This data can be compared to that of HDX on MFS sugar cation symporter Xyle, where substrate binding induces a transition to the OF state. There is no discussion of how this HDX data compares to previous MFS sugar transporter HDX. The manuscript could benefit from this comparison rather than a comparison to LacY. It is unlikely that there are universal mechanisms that can be inferred even from these model proteins. Highlighting differences between these transport systems provides broader insights into this protein class. Doi: 10.1021/jacs.2c06148 and 10.1038/s41467-018-06704-1.

The sugar translocation free-energy landscape simulations showed that both helix bundles move relative to the membrane plane. This analysis aimed to clarify a hypothesis in the field—that the MFS transporter can use an asymmetric mode to perform the conformational transition between inward- and outward-facing states. In the case of MelB_{St} , we clearly demonstrated that both domains move and each helix bundle moves as a unit. So only a small number of helices and loops showed labeling changes. Thanks for the suggestion about comparing with Xyle. We have included that in the discussion.

(10) Additionally, the recent publication of SMFS data (by the authors: doi:10.1016/j.str.2022.11.011) states the following: "In the presence of either melibiose or a coupling Na^+ -cation, however, MelB increasingly populates the mechanically less stable state which shows a destabilized middle-loop C3." And "In the presence of both substrate and co-substrate, this mechanically less stable state of MelB is predominant." It would benefit the authors to comment on these data in contrast to the HDX obtained here.

Additionally, is the C3 loop covered, and does it show the destabilization suggested by these studies? HDX can provide a plethora of results that are missing from the current analysis on ligand allostery. The authors instead chose to reference CD and thermal denaturation methods as comparisons.

Thank this reviewer for reading the single-molecule force spectroscopy (SMFS) study on MelB_{St}. The C3 loop mentioned in this SMFS article is partially covered in the dataset Mel or Mel plus Na⁺ vs. apo, and there is more coverage in the Na⁺ vs. apo dataset. In either condition, no deprotection was detected. The labeling time point might not be long enough to detect it.

Reviewer #3:

Summary:

The melibiose permease from Salmonella enterica serovar Typhimurium (MelB_{St}) is a member of the Major Facilitator Superfamily (MFS). It catalyzes the symport of a galactopyranoside with Na⁺, H⁺, or Li⁺, and serves as a prototype model system for investigating cation-coupled transport mechanisms. In cation-coupled symporters, a coupling cation typically moves down its electrochemical gradient to drive the uphill transport of a primary substrate; however, the precise role and molecular contribution of the cation in substrate binding and translocation remain unclear. In a prior study, the authors showed that the binding affinity for melibiose is increased in the presence of Na⁺ by about 8-fold, but the molecular basis for the cooperative mechanism remains unclear. The objective of this study was to better understand the allosteric coupling between the Na⁺ and melibiose binding sites. To verify the sugar-recognition specific determinants, the authors solved the outward-facing crystal structures of a uniport mutant D59C with four sugar ligands containing different numbers of monosaccharide units (α-NPG, melibiose, raffinose, or α-MG). The structure with α-NPG bound has improved resolution (2.7 Å) compared to a previously published structure and to those with other sugars. These structures show that the specificity is clearly directed toward the galactosyl moiety. However, the increased affinity for α-NPG involves its hydrophobic phenyl group, positioned at 4 Å-distance from the phenyl group of Tyr26, which forms a strong stacking interaction. Moreover, a water molecule bound to OH-4 in the structure with α-NPG was proposed to contribute to the sugar recognition and appears on the pathway between the two specificity-determining pockets. Next, the authors analyzed by hydrogen-to-deuterium exchange coupled to mass spectrometry (HDX-MS) the changes in structural dynamics of the transporter induced by melibiose, Na⁺, or both. The data support the conclusion that the binding of the coupling cation at a remote location stabilizes the sugar-binding residues to switch to a higher-affinity state. Therefore, the coupling cation in this symporter was proposed to be an allosteric activator.

Strengths:

- (1) The manuscript is generally well written.*
- (2) This study builds on the authors' accumulated knowledge of the melibiose permease and integrates structural and HDX-MS analyses to better understand the communication between the sodium ion and sugar binding sites. A high sequence coverage was obtained for the HDX-MS data (86-87%), which is high for a membrane protein.*

Thank this reviewer for your positive comments.

Weaknesses:

- (1) I am not sure that the resolution of the structure (2.7 Å) is sufficiently high to unambiguously establish the presence of a water molecule bound to OH-4 of the α-NPG*

sugar. In Figure 2, the density for water 1 is not obvious to me, although it is indeed plausible that water mediates the interaction between OH4/OH6 and the residues Q372 and T373.

A water molecule can be modeled at a resolution ranging from 2.4 to 3.2 Å, and the quality of the model depends on the map quality and water location. In this revision, we refined the resolution to 2.6 Å using the same dataset and also performed all-atom MD simulations. All results support the occupancy of water-1 in the sugar-bound MelB_{St}.

(2) Site-directed mutagenesis could help strengthen the conclusions of the authors. Would the mutation(s) of Q372 and/or T373 support the water hypothesis by decreasing the affinity for sugars? Mutations of Thr121, Arg 295, combined with functional and/or HDX-MS analyses, may also help support some of the claims of the authors regarding the allosteric communication between the two substrate-binding sites.

The authors thank this reviewer for the thoughtful suggestions. MelB_{St} has been subjected to Cys-scanning mutagenesis (<https://doi.org/10.1016/j.jbc.2021.101090>). Placing a Cys residue at Gln372 significantly decreased the transport initial rate, accumulation, and melibiose fermentation, with minimal effect on protein expression, as shown in Figure 2 of this JBC article, which could support its role in the binding pocket. The T373C mutant retained most of the WT's activities. Our previous studies showed that Thr121 is only responsible for Na⁺ binding in MelB_{St}, and mutations decreased protein stability; now, HDX reveals that this is the rigid position. Additionally, our previous studies indicated that Arg295 is another conformationally important residue. In this version, we have added more HDX analysis to explore the relationship between the two substrate-binding sites with conformational dynamics, especially focusing on the gating salt-bridge network including Arg295, which has provided meaningful new insights.

(3) The main conclusion of the authors is that the binding of the coupling cation stabilizes those dynamic sidechains in the sugar-binding pocket, leading to a high-affinity state. This is visible when comparing panels c and a from Figure S5. However, there is both increased protection (blue, near the sugar) and decreased protection in other areas (red). The latter was less commented, could the increased flexibility in these red regions facilitate the transition between inward- and outward-facing conformations? The HDX changes induced by the different ligands were compared to the apo form (see Figure S5). It might be worth it for data presentation to also analyze the deuterium uptake difference by comparing the conditions sodium ion+melibiose vs melibiose alone. It would make the effect of Na⁺ on the structural dynamics of the melibiose-bound transporter more visible. Similarly, the deuterium uptake difference between sodium ion+melibiose vs sodium ion alone could be analyzed too, in order to plot the effect of melibiose on the Na⁺-bound transporter.

Thanks for this important question. We have added more discussion of the deprotected data and prepared a new Fig. 8b to highlight the melibiose-binding-induced flexibility in several loops, especially the gating area on both sides of the membrane. We also proposed that these changes might facilitate the formation of the transition-competent state. The overall effects induced by substrate binding are relatively small, and the datasets for apo and Na were collected separately, so comparing melibiose&Na⁺ versus Na⁺ might not be as precise. In fact, the Na⁺ effects on the sugar-binding site can be clearly seen in the deuterium uptake plots shown in Figures 7-8, by comparing the first and last panels.

(4) For non-specialists, it would be beneficial to better introduce and explain the choice of using D59C for the structural analyses.

Asp59 is the only site that responds to the binding of all coupling cations: Na⁺, Li⁺, or H⁺. Notably, this thermostable mutant D59C selectively abolishes all cation binding and

associated cotransport activities, but it maintains intact sugar binding and exhibits conformational transition as the WT, as demonstrated by electroneutral transport reactions including α -NPG transport showed in this articles, and melibiose exchange and fermentation showed previously. Therefore, the structural data derived from this mutant are significant and offer important mechanistic insights into sugar transport, which supports the conclusion that the Na^+ functions as allosteric activator.

(5) In Figure 5a, deuterium changes are plotted as a function of peptide ID number. It is hardly informative without making it clearer which regions it corresponds to. Only one peptide is indicated (213-226). I would recommend indicating more of them in areas where deuterium changes are substantial.

We appreciate this comment and have modified the plots by marking the residue position as well as labeled several peptides of significant HDX in the Fig 5b. We also provided a deuteration map based on peptide coverage (Fig. 5a).

(6) From prior work of the authors, melibiose binding also substantially increases the affinity of the sodium ion. Can the authors interpret this observation based on the HDX data?

This is an intriguing mechanistic question. In this HDX study, we found that the cation-binding pocket and nearby sugar-binding residues are conformationally rigid, while some sugar-binding residues farther from the cation-binding pocket are flexible. We concluded that conformational dynamics regulate sugar-binding affinity, but the increase in Na-binding affinity caused by melibiose is not related to protein dynamics. Our previous interpretation based on structural data remains our preferred explanation; therefore, the bound melibiose physically prevents the release of Na^+ or Li^+ from the cation-binding pocket. We also proposed the mechanism of intracellular Na^+ release in the 2024 JBC paper (<https://doi.org/10.1016/j.jbc.2024.107427>); after sugar release, the rotamer change of Asp55 will help Na^+ exit the cation pocket into the empty sugar pocket, and the negative membrane potential inside the cell will further facilitate movement from MelB_{St} to the cytosol.

Recommendations for the authors:

Reviewing Editor Comments:

(1) It would help the reader if the previous work were introduced more clearly, and if the results of the experiments reported in this manuscript were put into the context of the previous work. Lines 283-296 discuss observations that are similar to previous reported structures as well as novel interpretations. It would help the reader to be clearer about what the new observations are.

Thank you for the important comment. We have revised accordingly by adding related citations and words “as showed previously” when we stated our previous observations.

(2) The affinity by ITC is measured for various ligands, but very few conclusions are drawn about how the affinity correlates with the binding modes. Are the other ligands that are investigated in this study transported by the protein, or do they just bind? Can the protein transport the trisaccharide raffinose? The authors comment that raffinose exhibiting poor binding affinity despite having more sugar units is surprising, but this is not surprising to me. No additional interactions can be mapped to these units on their structure, and while it fits into the substrate binding cavity, the extra bulk of additional sugar units is likely to reduce affinity. In fact, from their listed ITC measurements, this appears to be the trend.

Additionally, the D59C mutant utilized here in structural determination is deficient in sodium/cation binding. The reported allostery of sodium-sugar binding will likely

influence the sugar binding motif as represented by these structures. This is clearly represented by the authors' own ITC work. The ITC included in this work was carried out on the WT protein in the presence of Na⁺. The authors could benefit from clarifying how this work fits with the structural work or carrying out ITC with the D59C mutant, or additionally, in the absence of sodium. For non-specialists, please better introduce and explain the choice of using D59C for the structural analyses.

Thank you for the meaningful comments. We have comprehensively addressed all the concerns and suggestions as listed in the summary of this revision. Notably, the D59C mutant does not catalyze any electrogenic melibiose transport involved in a cation transduction but catalyze downhill transport location of the galactosides, as shown by the downhill α -NPG transport assay in Fig. 1a. The intact downhill transport results from D59C mutant further supports the allosteric coupling between the cation- and sugar-binding sites.

The binding isotherm and poor affinity of the ITC measurements do not support to further analyze the binding mode since none showed sigmoidal curve, so the enthalpy change cannot be accurately determined. But authors thank this comment.

(3) It is not clear what Figure 2 is comparing. The text suggests this figure is a comparison of the lower resolution structure to the structure presented in this work; however, the figure legend does not mention which is which, and both images include a modelled water molecule that was not assigned due to poor resolution previously, as stated by the authors, in the previously generated structure. This figure should be more clearly explained.

We have addressed these concerns in the response to the Public Reviews at reviewer-2 #1.

(4) I am not sure that the resolution of the structure (2.7 Å) is sufficiently high to unambiguously establish the presence of a water molecule bound to OH-4 of the α -NPG sugar. In Figure 2, the density for water 1 is not obvious to me, although it is indeed plausible that water mediates the interaction between OH4/OH6 and the residues Q372 and T373. Please change line 278 to state "this OH-4 water molecule is likely part of sugar binding".

We have addressed these concerns in the response to the Public Reviews at reviewer-3 #1.

(5) Line 290-296: The Thr121 is not represented in any figures, while the Lys377 is. Their relative positioning between sugar water and sodium is not made clear by any figure.

Thanks for this comment. This information has been clearly presented in the Figs. 7-8. Lys377 is closer to the cation site and related far from the sugar-binding site.

(6) Methodology includes a lipid removal step. Based on other included methods, I assumed that the HDX-MS was being carried out in detergent-solubilized protein samples. I therefore do not see the need for a lipid removal step that is usually included for bilayer reconstituted samples. I note that this methodology is the same as previously used for MelB. It should be clarified why this step was included, if it was in fact used, aka, further details on the sample preparation should be included.

(7) A summary of HDX conditions and results should be given as recommended, including the mean peptide length and average redundancy per state alongside other included information such as reaction temperature, sequence coverage, etc., as prepared for previous publications from the authors, i.e., Hariharan et al., 2024.

We have addressed these concerns in the response to the Public Reviews at reviewer-2 #4.

(8) Uptake plots per peptide for the HDX-MS data should be included as supporting information outside of the few examples given in Figure 6.

We have addressed these concerns in the response to the Public Reviews at reviewer-2 #4.

*(9) A reference should be given to the hybrid significance testing method utilised. Additionally, as stated by Hageman and Weis (2019) (doi:10.1021/acs.analchem.9b01325), the use of $P < 0.05$ greatly increases the likelihood of false positive ΔD identifications. While the authors include multiple levels of significance, what they refer to as high and lower significant results, and this reviewer understands that working with dynamic transporters can lead to increased data variation, a statement of why certain statistical criteria were chosen should be included, and possibly accompanied by volcano plots. The legend of Figure 6 should include what P value is meant by * and ** rather than statistically significant and highly statistically significant.*

We have addressed these concerns in the response to the Public Reviews at reviewer-2 #4.

(10) The table (S3) and figure (S4) showing uncovered residues is an unclear interpretation of the data; this would be better given as a peptide sequence coverage heat map. This would also be more informative for the redundancy in covered regions, too. In this way, S3 and S4 can be combined.

We have addressed these concerns in the response to the Public Reviews at reviewer-2 #4.

(11) Residual plots in Figure 5 could be improved by a topological map to indicate how peptide number resembles the protein amino acid sequence.

Thanks for the request, due to the figure 6 is big so that we add a transmembrane topology plot colored with the HDX results in Fig. 8c.

(12) The presentation of data in S5 could be clarified. Does the number of results given in the brackets indicate overlapping peptides? What are the lengths of each of these peptides? Classical HDX data presentation utilizes blue for protection and red for deprotection. The use of yellow ribbons to show protection in non-sugar binding residues takes some interpretation and could be clarified by also depicting in a different blue. I also don't see the need to include ribbon and cartoon representation when also using colors to depict protection and deprotection. The authors should change or clarify this choice.

We have moved this figure into the current Fig. 6b as suggested by Reviewer-3. To address your questions listed in the figure legend, the number of results shown in brackets indeed indicates overlapping peptides. What are the lengths of each of these peptides? The sequences of each peptide are shown in Figures 7-8 and are also included in Supplemental Figure S5. Regarding the use of color, both blue and green were used to distinguish peptides protecting the substrate-binding site from other regions. The ribbon and cartoon representations are provided for clarity, as the cartoon style hides many helices.

(13) In Table S5, the difference between valid points and protection is unclear. And what is indicated by numbers in brackets or slashes? Additionally, it should be highlighted again here that single-residue information is inferred from peptide-level data. By value, are the authors referring to peptide-level differential data?

Please review our responses in the Public Reviews at reviewer-2 #5.

(14) Line 316 states a significant difference in seen in dynamics, how is significance measured here? There is no S.D. given in Table S4. Can the authors further comment on the potential involvement in solvent accessibility and buried helices that might influence the overall dynamics outside of their role in sugar vs sodium binding? An expected low rate of exchange suggests that dynamics are likely influenced by solvent accessibility or

peptide hydrophobicity? The increased dynamics at peptides covering the Na binding site on overall more dynamic helices suggests that there isn't a difference between the dynamics of each site.

Please review our responses in the Public Reviews at reviewer-2 #5.

(15) Previously stated HDX-MS results of MelB (Hariharan et al., 2024) state that the transmembrane helices are less dynamic than polypeptide termini and loops with similar distributions across all transmembrane bundles. The previous data was obtained in the presence of sodium. Does this remove the difference in dynamics in the sugar-binding helices and the cation-binding helices? Including this comparison would support the statement that the sodium-bound MelB is more stable than the Apo state, along with the lack of deprotection observed in the differential analysis.

Please review our responses in the Public Reviews.

(16) MD of MelB suggests all transmembrane helices are reorientated during substrate translocation, yet substrate and cotransporter ligand binding only significantly impacts a small number of helices. Can the authors comment on the ensemble of states expected from each HDX experiment? The data presented here instead shows overall stabilisation of the transporter. This data can be compared to that of HDX on MFS sugar cation symporter XylE, where substrate binding induces a transition to the OF state. There is no discussion of how this HDX data compares to previous MFS sugar transporter HDX. The manuscript could benefit from this comparison rather than a comparison to LacY. It is unlikely that there are universal mechanisms that can be inferred even from these model proteins. Highlighting differences instead between these transport systems provides broader insights into this protein class. Doi: 10.1021/jacs.2c06148 and 10.1038/s41467-018-06704-1.

Please review our responses in the Public Reviews.

(17) Additionally, the recent publication of SMFS data (by the authors: doi:10.1016/j.str.2022.11.011) states the following: "In the presence of either melibiose or a coupling Na⁺-cation, however, MelB increasingly populates the mechanically less stable state which shows a destabilized middle-loop C3." And "In the presence of both substrate and co-substrate this mechanically less stable state of MelB is predominant." It would benefit the authors to comment on these data in contrast to the HDX obtained here. Additionally, is the C3 loop covered, and does it show the destabilization suggested by these studies? HDX can provide a plethora of results that are missing from the current analysis on ligand allostery. The authors instead chose to reference CD and thermal denaturation methods as comparisons.

Please review our responses in the Public Reviews.

(18) The main conclusion of the authors is that the binding of the coupling cation stabilizes those dynamic sidechains in the sugar-binding pocket, leading to a high-affinity state. This is visible when comparing panels c and a from Figure S5. However, there is both increased protection (blue, near the sugar) and decreased protection in other areas (red). The latter was less commented, could the increased flexibility in these red regions facilitate the transition between inward- and outward-facing conformations? The HDX changes induced by the different ligands were compared to the apo form (see Figure S5). It might be worth it for data presentation more visible to also analyze the deuterium uptake difference by comparing the conditions sodium ion+melibiose vs melibiose alone. You would make the effect of Na⁺ on the structural dynamics of the melibiose-bound transporter. Similarly, the deuterium uptake difference between sodium ion+melibiose vs sodium ion alone could be analyzed too, in order to plot the effect of melibiose on the Na⁺-bound transporter.

Please review our responses in the Public Reviews.

(19) In Figure 5a, deuterium changes are plotted as a function of peptide ID number. It is hardly informative without making it clearer which regions it corresponds to. Only one peptide is indicated (213-226); I would recommend indicating more of them, in areas where deuterium changes are substantial.

Please review our responses in the Public Reviews.

(20) Figure 6, please indicate in the legend what the black and blue lines are (I assume black is for the apo?)

We are sorry that we did not make it clear. Yes, the black was used for apo state and blue was used for all bound states

(21) From prior work of the authors, melibiose binding also substantially increases the affinity of the sodium ion. Can the authors interpret this observation based on the HDX data?

Please review our responses in the Public Reviews.

Addressing the following three points would strengthen the manuscript, but also involve a significant amount of additional experimental work. If the authors decide not to carry out the experiments described below, they can still improve the assessment by focusing on points (1-21) described above.

(22) Have the authors considered carrying out an HDX-MS comparison between the WT and the D59C mutant? This may provide some further information on the WT structure (particularly a comparison with sugar-bound). This could be tied into a nice discussion of their structural data.

Please review our responses in the Public Reviews.

(23) Have the authors considered utilising Li^+ to infer how cation selectivity impacts the allostery? Do they expect similar stabilisation of a higher-affinity sugar binding state with all cations?

Please review our responses in the Public Reviews.

(24) Site-directed mutagenesis could help strengthen the conclusions. Would the mutation(s) of Q372 and/or T373 support the water hypothesis by decreasing the affinity for sugars? Mutations of Thr 121 and Arg 295, combined with functional and/or HDX-MS analyses, may also help support some of the authors' claims regarding allosteric communication between the two substrate-binding sites.

Please review our responses in the Public Reviews.

<https://doi.org/10.7554/eLife.108335.2.sa0>